AN ADVANCED UNDERGRADUATE AND GRADUATE INTRODUCTION

Introduction to Complex Relativity

From clocks and light cones to
complex geometry and physical dynamics

$$\gamma = g + \mathrm{i}b \quad \longrightarrow \quad \gamma = (1 + \mathrm{i}\chi)g$$

$$g(v, v) = 0 \implies \gamma(v, v) = 0$$

A guided account of the CR research framework, with explicit derivations, worked examples, geometric illustrations, exercises, and selected solutions.

The reader needs special relativity, linear algebra, and Lagrangian mechanics. Differential geometry, constrained dynamics, and quantum field theory enter in stages. The complex numbers arrive early; the difficult questions are allowed to arrive on time.

Charles K

3 November 2026

Before we begin

These notes teach the specific *Complex Relativity* (CR) framework developed in the accompanying research manuscripts. Their immediate source is the revised English synthesis of the P1/P2 formulation, including its main-text treatment of response-target curvature.[1] The aim is to make that framework intelligible, calculable, and open to criticism. This is an introduction to a research program, not a claim that its interacting models or proposed quantum completion have already been established experimentally.

The distinction matters pedagogically. A student should be able to locate the exact line at which a theorem ends and a model begins. The common-null-cone theorem below is elementary and exact. The restricted gravitational action is an additional dynamical choice. The response family is another such choice. Its positive sectors have a well-defined geometry, but assigning them to observed particles requires still more structure. Keeping these steps visible makes the theory more interesting, because the unsolved questions become questions one can actually work on.

Our route starts with real measurements. Clocks tick on real worldlines, light signals identify null directions, and a physical momentum pairs with a real displacement to give a real action increment. Complex geometry must respect these facts. Once it does, its remaining scalar freedom can be given dynamics and coupled to matter. The later chapters explore what this construction does, and what it does not yet say about quantum physics.

There are three useful ways through the notes.

- **First pass:** read Sections 1–8, the explanations surrounding the curvature calculation in Section 10, and the cosmology and concluding sections. Skip the paragraphs marked *Graduate extension* on this pass.
- **Course pass:** work through the main derivations and the exercises in order. A compact course could use twelve meetings: two on foundations, two on particle and field dynamics, three on the response sectors, two on gauge and cosmological models, and three on records and quantum structure.
- **Research pass:** concentrate on the assumptions behind Sections 11, 13, 16, and 19. Reproduce the counterexamples as carefully as the constructive results.

Selected exercise solutions are included at the end. A problem left unsolved in the source research is identified as a research problem, rather than disguised as a homework exercise with an implausibly short answer.

Conventions that prevent avoidable arguments

We use signature $(-+++)$ and, after the first section, units $c_{\mathrm{light}} = \hbar = 1$ unless stated otherwise. The symbol c without a subscript is a *dimensionless response parameter*, never the speed of light. The reduced Planck mass satisfies $M_{\mathrm{P}}^{-2} = 8\pi G_{\mathrm{N}}$ in natural units. Greek indices refer to four-dimensional spacetime; Latin target indices refer to fields. Cosmic proper time is τ.

The physical metric is g. A complex spacetime metric is G, and its pullback to physical spacetime is γ. A response matrix is G_c. These objects have different jobs even when their letters look related. We write

$$F = 1 + \mathrm{i}\chi, \qquad \rho = |F| = \sqrt{1+\chi^2}, \qquad \phi = f\,\operatorname{arsinh}\chi, \qquad x = \phi/f.$$

Here χ is dimensionless; the canonically normalized geometric scalar ϕ and the constant f have mass dimension one. The scalar φ used in the free-field examples is a separate matter field. An energy density will carry a subscript, such as ρ_ϕ, and a quantum density operator will be denoted ϱ.

Our curvature convention gives a sphere positive scalar curvature. The canonical particle convention is $\Theta = p_\mu \mathrm{d}x^\mu$, $\omega = \mathrm{d}\Theta$, and $\iota_{X_H}\omega = -\mathrm{d}H$, which gives the usual Hamilton equations. Overall symplectic signs differ across textbooks; changing one sign is harmless, changing only half of a convention is not.

Contents

Before we begin 1

I Real measurements, complex geometry 3

1 Why put complex numbers into relativity? 3

2 Complex spacetime without eight physical directions 4

3 P1 and the common-null-cone theorem 5

4 P2: momentum, mass, and one real constraint 7

II How the geometry moves things 9

5 Real canonical dynamics 9

6 Fields, wavefronts, and the order of variation 10

7 The geometric scalar from a restricted curvature action 12

III The response sectors 15

8 A two-component response and all real values of c 15

9 Canonical fields and the shared scalar interaction 17

10 Target-space curvature of normal response 18

11 The critical sector: a zero mode is not automatically a gauge mode 21

IV Building physical models 24

12 One action, one shared source 24

13 Internal gauge structure: what the extra assumptions buy 25

14 Cosmology: what the simplest scalar actually does 27

15 A dark-sector example, including the decay nobody invited 28

V Quantum structure and physical observation 30

16 How much quantum mechanics is already in the geometry? 30

17 Probabilities need preparation and recording dynamics 32

18 An apparatus made from the fields 34

19 What would count as progress? 35

A A compact mathematical toolkit 37

B Exercise solutions and checks 38

C Notation, source map, and further reading 42

References 43

Part I

Real measurements, complex geometry

Begin with the things a laboratory can read, and ask what a complex metric is allowed to do.

1 Why put complex numbers into relativity?

1.1 Three uses of a complex number

A complex number in physics does not automatically represent an extra observable dimension. In alternating-current theory, $e^{\mathrm{i}\omega t}$ keeps track of a phase; the voltage is real. In quantum mechanics, complex amplitudes belong to a state space and yield probabilities through an additional rule. In complex geometry, coordinates and bilinear forms are complex-valued mathematical objects. These uses can communicate, but they are not interchangeable.

CR asks whether a complex spacetime geometry, constrained by real clocks and causal signals, can organize additional physical structure. It does not begin by declaring that ordinary time has become imaginary. Nor does it identify a complex spacetime coordinate with a quantum amplitude. Such identifications would be new hypotheses, each with its own obligations. The letter i is versatile, but it does not come with a complete theory of measurement attached.

The starting question is precise. Suppose the metric induced on real physical spacetime is

$$\gamma_{\mu\nu} = g_{\mu\nu} + \mathrm{i}b_{\mu\nu}, \tag{1.1}$$

where g is a real Lorentz metric and b is real and symmetric. What restrictions on b follow if every real lightlike direction is also null for the *full* complex interval? The answer will be $b = \chi g$. Ten apparently independent imaginary metric components collapse to one real scalar function. That reduction is the first substantial payoff.

1.2 A short return to special relativity

In coordinates $x^0 = c_{\text{light}}t$, Minkowski spacetime has

$$\mathrm{d}s^2 = -(\mathrm{d}x^0)^2 + (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2. \tag{1.2}$$

A timelike clock measures

$$\mathrm{d}\tau = \frac{1}{c_{\text{light}}}\sqrt{-g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu}. \tag{1.3}$$

For a worldline parameterized by τ, its four-velocity obeys $g(u,u) = -c_{\text{light}}^2$. A null signal instead satisfies $g(k,k) = 0$. Its proper-time increment is zero, so it cannot be assigned the same unit timelike normalization. A photon does not carry a wristwatch with a particularly small battery.

At a point, a Lorentz transformation preserves $g(v,v)$ and hence the null cone. In curved spacetime one can still choose a local inertial frame at any event. One cannot generally choose a single inertial frame covering an extended curved region: tidal effects survive. The distinction between pointwise kinematics and spacetime dynamics will recur throughout CR.

1.3 A cone is not a clock

If $\widetilde{g} = \Omega^2 g$ with $\Omega > 0$, then g and $\widetilde{g}$ have the same null directions. They do not give the same proper time: $\mathrm{d}\widetilde{\tau} = \Omega \mathrm{d}\tau$. Thus the null cone fixes a conformal class, while a clock calibration supplies a scale.

Radar measurements illustrate the division of labor. An observer emits a signal at proper time τ_-, receives its reflection at τ_+, and assigns

$$T = \frac{\tau_+ + \tau_-}{2}, \qquad R = \frac{c_{\text{light}}}{2}(\tau_+ - \tau_-). \tag{1.4}$$

Directions, nearby events, and a suitable propagation model are also needed to reconstruct a local geometry. One round-trip time is not ten independent metric components in disguise.

At fixed two-way speed, one-way coordinate speeds can change with a synchronization convention. For example, $c_+ = c_{\text{light}}/(1+\epsilon)$ and $c_- = c_{\text{light}}/(1-\epsilon)$ give the same round-trip time $2L/c_{\text{light}}$ when $|\epsilon| < 1$. Such a coordinate choice does not establish a new physical anisotropy. CR's complex compatibility condition will be an explicit premise, not a conclusion drawn from a choice of clock synchronization.

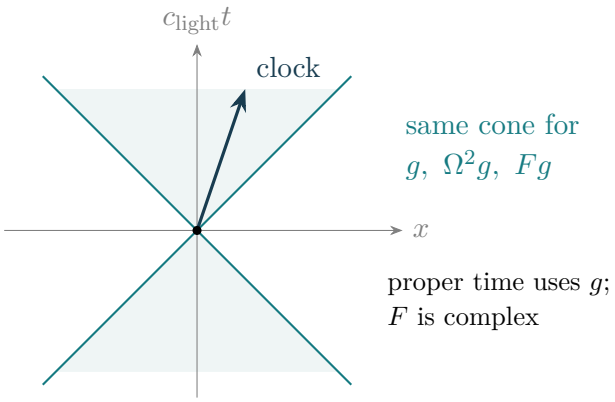


Figure 1: A two-dimensional section of the real null cone. Multiplication by a nonzero scalar preserves its zero set, including when that scalar is complex. Only the chosen real Lorentz metric defines the clock reading.

Exercise 1. Cone versus scale Let $\widetilde{g} = 4g$. Compare the null cones and the proper time along the same timelike displacement. Then explain why preserving all light rays is not, by itself, enough to determine the readings of massive clocks.

2 Complex spacetime without eight physical directions

2.1 The ambient space and the real slice

Take a complex manifold $\mathcal{Z}$ of complex dimension four. Locally it has coordinates $z^A = x^A + \mathrm{i}y^A$. As a real manifold it has dimension eight, but physical events are assumed to lie on a real four-dimensional embedded manifold M, with embedding $\iota : M \hookrightarrow \mathcal{Z}$. The real slice need not be globally given by $y^A = 0$; that is a convenient local picture.

The slice is *maximally real*: no nonzero tangent vector to it becomes another tangent vector to it after multiplication by i. In symbols,

$$\mathrm{d}\iota(T_pM) \cap J\mathrm{d}\iota(T_pM) = \{0\}, \qquad T_pM \otimes_{\mathbb{R}} \mathbb{C} \simeq T^{1,0}_{\iota(p)}\mathcal{Z}. \tag{2.1}$$

The second relation says that real physical tangent vectors form a basis for the complex tangent space once complex coefficients are allowed. It is this relation, rather than an unspecified instruction to complexify everything, that will fix the natural extension of momentum.

Let G be a nondegenerate symmetric complex-bilinear metric on $\mathcal{Z}$. Its pullback is $\gamma = \iota^* G$. We choose its real part $g = \Re\gamma$ to be the physical Lorentz metric. The imaginary part is b in Eq. (1.1). At this stage b is not assumed conformal.

2.2 Bilinear is not Hermitian

For a complex vector z, a bilinear form involves $z^A z^B$, whereas a Hermitian form involves $\bar{z}^A z^B$. Even in one dimension,

$$B(z,z) = z^2, \qquad h(z,z) = \bar{z}z = |z|^2$$

are different objects. A positive Hermitian norm has no nonzero null vectors. A Lorentzian complex-bilinear form does. Replacing one by the other halfway through a calculation would remove the very null geometry that CR is trying to preserve.

Hermitian structures will enter later on internal fibers and in one term of the response action. This is legitimate because those structures are introduced explicitly. There is no requirement that every appearance of a complex number use the same type of quadratic form.

Worked example 2.1 (A nontrivial complex factor)**.** On a local real slice let $g = \eta$, $\chi = 2$, and $G_{AB} = (1 + 2\mathrm{i})\eta_{AB}$. The pullback has $\Re\gamma = \eta$ and the same real null cone as η. Nevertheless $\gamma(v, v)$ is complex for a generic timelike v. The model is not inconsistent: the clock interval is defined by g, while the full complex interval obeys the additional compatibility condition on null directions. Equating every complex interval directly to a clock reading would be a different proposal.

2.3 Local complexification and its limits

Real-analytic functions admit local holomorphic extensions. Thus real-analytic g and χ can, locally and with nondegeneracy maintained, produce a compatible complex metric. An arbitrary smooth function need not admit a holomorphic extension. A local real effective field theory and a global holomorphic spacetime are therefore different levels of construction.

Similarly, one should not require the physical slice to be the fixed set of an *isometric* complex-conjugation involution unless one wants to force its induced metric to be real. For tangent vectors fixed by such an involution, metric compatibility would imply $\gamma = \bar{\gamma}$, hence $b = 0$. The present framework uses a real readout $g = \Re\gamma$ and allows $b \neq 0$.

> **Graduate extension.** The global problem includes compatible analytic charts, the existence of the real slice, nondegenerate extension of G, topology, and branches of the complex volume element. None is solved by writing a local coordinate $z = x + \mathrm{i}y$. The pointwise rigidity theorem in the next section does not need a global extension; a global fundamental theory would.

3 P1 and the common-null-cone theorem

3.1 What P1 actually asks for

The first postulate has an operational part and a geometric compatibility part. In the strengthened P1/P2 formulation used here:

1. Physical observations are made by clock-carrying observers on real timelike worldlines. Local inertial descriptions have the same free kinematic form under the appropriate Lorentz transformations. An observer may accelerate; an instantaneously comoving inertial frame is still available at each event.

2. A sufficiently rich family of signals and calibrations supplies the real null cone and clock scale. Every nonzero real null vector must annihilate the full complex interval:

$$g(v,v) = 0 \quad \Longrightarrow \quad \gamma(v,v) = 0. \tag{3.1}$$

3. Free motion has the ordinary inertial limit. Transport and evolution away from that limit must be specified by compatible real dynamics.

Complex spacetime and the null cone are the two mathematical and physical foundations. P1 is the statement linking them to observers. This avoids renaming the foundations themselves as two independent postulates.

3.2 A proof you can do on one blackboard

Theorem 3.1 (Null-cone rigidity)**.** *Let g be a Lorentz metric in four real dimensions and let b be a real symmetric bilinear form. If $b(v,v) = 0$ for every real g-null vector, then $b = \chi g$ for a real scalar χ.*

Proof. Choose an orthonormal frame with $g = \mathrm{diag}(-1,1,1,1)$. Every vector $(1,\boldsymbol{n})$ with $|\boldsymbol{n}| = 1$ is null. Therefore

$$b_{00} + 2b_{0i}n_i + b_{ij}n_i n_j = 0. \tag{3.2}$$

Subtract the equation evaluated at $-\boldsymbol{n}$. The result $4b_{0i}n_i = 0$ for every unit $\boldsymbol{n}$ implies $b_{0i} = 0$. Choosing $\boldsymbol{n} = e_i$ next gives $b_{ii} = -b_{00}$ for each spatial index. Finally choose $\boldsymbol{n} = (e_i + e_j)/\sqrt{2}$ for $i \neq j$. The already fixed diagonal terms cancel, leaving $b_{ij} = 0$. Thus

$$b = \mathrm{diag}(b_{00}, -b_{00}, -b_{00}, -b_{00}) = -b_{00}g.$$

Set $\chi = -b_{00}$. In coordinate-independent form $\chi = \frac{1}{4}g^{\mu\nu}b_{\mu\nu}$. □

Applying the theorem to the imaginary part of Eq. (3.1) gives

$$\boxed{\gamma = Fg, \qquad F = 1 + \mathrm{i}\chi.} \tag{3.3}$$

Since χ is real, F never vanishes on the physical slice. Conversely, Eq. (3.3) plainly satisfies Eq. (3.1). One-way inclusion of real null directions was already enough; a second, stronger cone assumption was not needed.

3.3 What was eliminated, and what survived?

A symmetric four-by-four matrix has ten independent components. The condition on the sphere in Eq. (3.2) removes nine independent combinations and leaves the conformal direction. Equivalently, the angular function contains one monopole, three dipoles, and five traceless quadrupoles. Setting all nine to zero leaves χ invisible to this null-direction test.

That invisibility has a physical interpretation. A cone can tell us whether an imaginary tensor distorts null directions relative to the real geometry. It cannot measure a scalar multiplying the entire metric, because zero multiplied by a scalar remains zero. Clock calibration fixes the scale of g; it does not automatically determine the dynamics of χ.

Worked example 3.2 (Covariance without cone compatibility)**.** Let $g = \eta$ and $b = \mathrm{diag}(\alpha, 0, 0, 0)$ with $\alpha \neq 0$. The pair (g, b) is perfectly capable of being written in covariant tensor notation. But for $v = (1, 1, 0, 0)$, $g(v, v) = 0$ and $\gamma(v, v) = \mathrm{i}\alpha \neq 0$. Coordinate covariance has not enforced P1.

Passive coordinate changes rewrite the same background. Active Lorentz symmetry asks whether that background is unchanged by a physical symmetry operation. Confusing these statements is a reliable way to derive a theorem whose crucial assumption has quietly disappeared.

Exercise 2. Nine null tests Impose Eq. (3.2) only at $\boldsymbol{n} = \pm e_1, \pm e_2, \pm e_3$ and $(e_1+e_2)/\sqrt{2}, (e_1+e_3)/\sqrt{2}, (e_2+e_3)/\sqrt{2}$. Show that these nine exact tests force $b = \chi g$ within the space of symmetric quadratic forms. Why would the same statement not follow for an arbitrary nonquadratic direction-dependent function?

4 P2: momentum, mass, and one real constraint

4.1 Start with what an observer measures

A physical momentum is a real covector $p \in T_x^*M$. An observer with four-velocity u measures

$$E = -p_\mu u^\mu. \tag{4.1}$$

This pairing is invariant under coordinate changes. With the chosen time orientation, it selects the physical positive-energy branch. It is more reliable than taking the real part of a particular coordinate component and calling that component the energy.

For each free species, P2 specifies a nonnegative real mass m and the real shell

$$C_{\mathrm{R}} = g^{\mu\nu} p_\mu p_\nu + m^2 = 0. \tag{4.2}$$

It also specifies natural complex momentum, compatibility of the full complex shell with this same real shell, and a real canonical realization of free dynamics. These clauses are substantive. A polynomial equation in a complex momentum, by itself, does not tell us which momenta are physical or how they evolve.

4.2 The natural extension

Because the slice is maximally real, every complex tangent vector has a unique form $v + \mathrm{i}w$ with real v, w. Define

$$p^{\mathbb{C}}(v + \mathrm{i}w) = p(v) + \mathrm{i}p(w). \tag{4.3}$$

This is the unique complex-linear extension of p. In particular, $p^{\mathbb{C}}(v) = p(v)$ on real displacements. We use *Scheme A*: $P = p^{\mathbb{C}}$.

The inverse of $\gamma = Fg$ is $F^{-1}g^{-1}$, so

$$G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) = F^{-1} g^{\mu\nu} p_\mu p_\nu. \tag{4.4}$$

Try the quadratic shell $G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + A(x)m^2 = 0$, with a momentum-independent coefficient A. For $m > 0$, requiring the entire physical shell Eq. (4.2) to satisfy it gives $A = F^{-1}$. Thus

$$\boxed{C_{\mathrm{A}} = G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + F^{-1}m^2 = F^{-1} C_{\mathrm{R}}.} \tag{4.5}$$

The complex coefficient is part of the representation of a real mass. It does not introduce a second independently measured mass parameter. Uniqueness here is only within the stated quadratic class. Multiplying a constraint by a nonzero function preserves its zero set, and more general off-shell constructions are possible.

4.3 Why a complex equation need not mean two constraints

Writing out the real and imaginary parts gives

$$\Re C_{\mathrm{A}} = \frac{C_{\mathrm{R}}}{1+\chi^2}, \qquad \Im C_{\mathrm{A}} = -\frac{\chi C_{\mathrm{R}}}{1+\chi^2}. \tag{4.6}$$

They are dependent. On the shell their gradients are collinear, so the constraint map has real rank one wherever $\mathrm{d}C_{\mathrm{R}} \neq 0$. The massive shell is regular; the zero-momentum vertex of the massless shell is excluded from this regular statement.

The physical phase space initially has eight real coordinates (x^μ, p_μ). One first-class mass-shell constraint and its reparameterization freedom remove two dimensions, leaving six. There is no extra removal merely because Eq. (4.5) was written with a complex coefficient.

Worked example 4.1 (A numerical shell check)**.** Set $\chi = 2$, $m = 3$, and spatial momentum $\boldsymbol{p} = (4, 0, 0)$ in a local inertial frame. Then $E = 5$ and $C_{\mathrm{R}} = -25 + 16 + 9 = 0$. Since $F^{-1} = (1 - 2\mathrm{i})/5$, both parts of C_{A} vanish together. If instead $E = 6$, then $C_{\mathrm{R}} = -11$ and $C_{\mathrm{A}} = -11/5 + 22\mathrm{i}/5$. The two nonzero components still encode the same single residual, not two independently adjustable errors.

4.4 Two tempting shortcuts

First, retain natural momentum but leave the mass coefficient equal to one:

$$F^{-1} g^{-1}(p,p) + m^2 = 0 \quad \Longleftrightarrow \quad g^{-1}(p,p) + (1+\mathrm{i}\chi) m^2 = 0.$$

For real p, m, the imaginary part demands $\chi m^2 = 0$. This excludes massive particles whenever $\chi \neq 0$. The compatible coefficient in Eq. (4.5) is doing necessary work.

Second, choose $P_B = \sqrt{F}\, p^{\mathbb{C}}$. Then the literal complex expression $G^{-1}(P_B, P_B) + m^2$ equals C_{R}. This is a useful alternative, *Scheme B*, but it rotates the momentum reality condition. Moreover, the canonical one-form becomes $\sqrt{F}\, p_\mu \mathrm{d}x^\mu$, whose exterior derivative contains a term involving $\mathrm{d}\sqrt{F}$. To match Scheme A's dynamics, one must calibrate the physical one-form by $F^{-1/2}$. Matching a shell is easier than matching an entire Hamiltonian system.

For real χ, F lies in the right half-plane, so a smooth principal square root exists on the real slice. The issue is therefore physical identification and canonical structure, not a local square-root crisis.

Exercise 3. The rank of a constraint For a real function C and a real field χ, calculate $\mathrm{d}\Re[(1+\mathrm{i}\chi)^{-1} C]$ and $\mathrm{d}\Im[(1+\mathrm{i}\chi)^{-1} C]$. Show explicitly why the two gradients become dependent at $C = 0$, even though their off-shell derivatives contain different $\mathrm{d}\chi$ terms.

> **Graduate extension.** A related rigidity result can run in the opposite direction. Suppose a fixed complex quadratic form h equals $-Am^2$ on an open patch of a positive-mass shell, with A independent of momentum direction. Then $K = h - Ag^{-1}$ vanishes there. Rescaling timelike covectors extends the zero set to an open cone. A quadratic polynomial vanishing on an open set is zero, so $h = Ag^{-1}$. Direction independence and the fixed quadratic class are essential; allowing an arbitrary compensating $A(p)$ destroys the conclusion.

Part II

How the geometry moves things

A mass shell says which momenta are allowed. An action says what happens next.

5 Real canonical dynamics

5.1 Pulling back the canonical one-form

On a cotangent bundle the canonical one-form is especially simple: $\Theta_{\mathbb{C}} = P_A \mathrm{d}z^A$. It pairs momentum with displacement. On the physical slice, natural complexification gives

$$j^*\Theta_{\mathbb{C}} = p_\mu \mathrm{d}x^\mu = \Theta_{\mathbb{R}}, \qquad \omega_{\mathbb{R}} = \mathrm{d}p_\mu \wedge \mathrm{d}x^\mu. \tag{5.1}$$

Here j is the embedding of real cotangent phase space induced by Eq. (4.3). This is a reason to prefer Scheme A for the initial development: the physical pairing is real without an additional rotation of momentum or a position-dependent calibration.

Use the real representative $\Re(FC_{\mathrm{A}}) = C_{\mathrm{R}}$ and introduce a real multiplier $e(\sigma)$:

$$S_{\mathrm{p}} = \int \mathrm{d}\sigma \left[p_\mu \dot{x}^\mu - \frac{e}{2}(g^{\mu\nu} p_\mu p_\nu + m^2) \right]. \tag{5.2}$$

A dot denotes differentiation with respect to an arbitrary worldline parameter σ. The multiplier is often called an einbein. Its purpose is not to add a new particle mode; its variation imposes the mass-shell constraint.

5.2 Deriving the trajectory

Independent variations of p_μ, x^μ, and e give

$$\dot{x}^\mu = e g^{\mu\nu} p_\nu, \tag{5.3}$$

$$\dot{p}_\mu = -\frac{e}{2} \partial_\mu g^{\alpha\beta} p_\alpha p_\beta, \qquad C_{\mathrm{R}} = 0. \tag{5.4}$$

On a patch with $e \neq 0$, eliminate $p_\mu = e^{-1} g_{\mu\nu} \dot{x}^\nu$:

$$S_{\mathrm{p}} = \frac{1}{2} \int \mathrm{d}\sigma \left[e^{-1} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu - e m^2 \right]. \tag{5.5}$$

For a massive particle on the future timelike branch, the positive solution of the multiplier equation is $e = \sqrt{-g(\dot{x}, \dot{x})}/m$. Substitution gives

$$S_m = -m \int \sqrt{-g_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu} = -m \int \mathrm{d}\tau. \tag{5.6}$$

Restoring units changes the last expression to $-m c_{\mathrm{light}}^2 \int \mathrm{d}\tau$. In proper-time gauge, $e = 1/m$ and $p^\mu = m u^\mu$.

The second-order equation obtained from Eq. (5.5) is

$$\ddot{x}^\mu + \Gamma(g)^\mu{}_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = \frac{\dot{e}}{e} \dot{x}^\mu. \tag{5.7}$$

For constant e this is the affine geodesic equation. The term on the right in a general parameter is tangent to the same curve; it changes the parameterization, not the path.

For $m = 0$, keep Eq. (5.5). The multiplier imposes a null trajectory and can be used to choose an affine parameter. The massive gauge $e = 1/m$ is not a sensible massless limit. Taking a limit of a bad coordinate choice is a common source of excellent-looking infinities.

5.3 Which connection transports a clock?

The complex and real Levi–Civita connections differ by

$$\Gamma(\gamma)^{\mu}{}_{\nu\lambda} - \Gamma(g)^{\mu}{}_{\nu\lambda} = \frac{1}{2}\left(\delta^{\mu}_{\nu}\partial_{\lambda}\ln F + \delta^{\mu}_{\lambda}\partial_{\nu}\ln F - g_{\nu\lambda}\nabla^{\mu}\ln F\right). \tag{5.8}$$

Along a tangent u, this difference contributes

$$(u^{\nu}\partial_{\nu}\ln F)u^{\mu} - \frac{1}{2}g(u,u)\nabla^{\mu}\ln F. \tag{5.9}$$

For timelike motion the second term is generally transverse. Thus a real clock geodesic of g is not automatically a complex geodesic of γ. For null motion the transverse term vanishes, but the remaining coefficient can be complex; removing it might require a complex reparameterization. Physical clock and affine parameters are fixed by the real action, not by this formal observation alone.

At fixed g and m, Eq. (5.2) contains no χ. Therefore the reference free particle has no direct scalar force. This does not make χ unphysical: it can gravitate, and additional response-dependent masses or interactions can couple it to matter. It simply tells us what this particular free action predicts.

Exercise 4. A position-dependent mass Replace m by $m(\phi)$ in S_m. In proper-time parameterization, show that

$$u^{\nu}\nabla_{\nu}u^{\mu} = -(g^{\mu\nu} + u^{\mu}u^{\nu})\nabla_{\nu}\ln m.$$

Explain why the force is orthogonal to u, and identify the assumption that made it absent in the reference free-particle model.

6 Fields, wavefronts, and the order of variation

6.1 A massive wave can have a null wavefront

The real Klein–Gordon equation, with $\hbar$ temporarily restored, is

$$\left(\Box_g - \frac{m^2}{\hbar^2}\right)\varphi = 0, \qquad \Box_g = \frac{1}{\sqrt{-g}}\partial_{\mu}\left(\sqrt{-g}g^{\mu\nu}\partial_{\nu}\right). \tag{6.1}$$

For a local WKB expression $\varphi = a\,e^{\mathrm{i}S/\hbar}$, the leading terms give

$$g^{\mu\nu}\partial_{\mu}S\,\partial_{\nu}S + m^2 = 0. \tag{6.2}$$

This is the Hamilton–Jacobi mass shell. The complex WKB notation is a calculation device; a real solution can be recovered by taking a real linear combination.

The *principal symbol* keeps only the highest derivatives in the PDE. It is

$$P(k) = g^{\mu\nu}k_{\mu}k_{\nu}. \tag{6.3}$$

Its zero set is null, even for a massive field. There is no contradiction. In flat space, $\omega^2 = |\boldsymbol{k}|^2 + m^2$, so a narrow packet has group speed $|\boldsymbol{k}|/\omega < 1$, while the high-frequency wavefront approaches the light cone. A packet trajectory, a dispersion relation, and the boundary of causal propagation answer related but different questions.

Worked example 6.1 (A slow packet and a lightlike front)**.** For $|\boldsymbol{k}| = m$, the group velocity is $1/\sqrt{2}$. For $|\boldsymbol{k}| = 10m$ it is $10/\sqrt{101}$. Both packets solve an equation with the same principal cone $-\omega^2 + |\boldsymbol{k}|^2 = 0$. The mass changes the dispersion but not the highest-derivative causal cone.

6.2 The naive complex equation adds an unwanted condition

In four spacetime dimensions,

$$\sqrt{-\gamma} = F^2\sqrt{-g}, \qquad \Box_\gamma \varphi = F^{-1}\left[\Box_g \varphi + \nabla^\mu \ln F\, \partial_\mu \varphi\right]. \tag{6.4}$$

The volume branch is chosen continuously from $F = 1$. It is the complex volume $F^2\sqrt{-g}$, not $|F|^2\sqrt{-g}$.

One might try $(\Box_\gamma - F^{-1}m^2)\varphi = 0$. Multiplying by F gives

$$(\Box_g - m^2)\varphi + \nabla^\mu \ln F\, \partial_\mu \varphi = 0. \tag{6.5}$$

If the physical matter field φ is real, the imaginary part requires

$$\nabla^\mu \chi\, \partial_\mu \varphi = 0, \tag{6.6}$$

because $\Im(\mathrm{d}\ln F) = \mathrm{d}\chi/(1+\chi^2)$. On a test background $g = \eta$, $\chi = \kappa t$ with $\kappa \neq 0$, this forces $\partial_t \varphi = 0$. Generic free oscillations have vanished from the theory. The background need not solve all gravitational equations for this operator-level counterexample to make its point: matching an algebraic shell did not determine a suitable wave equation.

6.3 A real action that works exactly

Instead, specify real fields and allowed variations first, impose $\gamma = Fg$, and choose

$$S_0 = -\frac{1}{2}\Re \int \mathrm{d}^4x \sqrt{-\gamma}\left[\gamma^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi + F^{-1}m^2\varphi^2\right]. \tag{6.7}$$

The volume supplies F^2, while each term in the bracket supplies F^{-1}. The integrand becomes F times a real expression. Since $\Re F = 1$,

$$S_0 = -\frac{1}{2}\int \mathrm{d}^4x \sqrt{-g}\left[(\nabla\varphi)^2 + m^2\varphi^2\right]. \tag{6.8}$$

This is an off-shell identity of actions. Varying the real field gives the usual real Klein–Gordon equation without Eq. (6.6).

This example teaches a general lesson. Restricting the configuration space and then varying is not the same operation as varying a larger complex theory and imposing restrictions afterward. The larger theory has equations in directions that the restricted theory has deliberately removed. Those equations need not be implied by the restricted ones.

6.4 Compatible actions are not unique

For any positive real function $Z(\chi)$, the action

$$S_Z = -\frac{1}{2}\int \mathrm{d}^4x \sqrt{-g}\, Z(\chi)\left[(\nabla\varphi)^2 + m^2\varphi^2\right] \tag{6.9}$$

has principal symbol $Zg^{\mu\nu}k_\mu k_\nu$ and the same leading WKB mass shell. Its equation, however, includes $\nabla \ln Z \cdot \nabla\varphi$, which changes amplitude transport. After the field redefinition $\psi = \sqrt{Z}\,\varphi$, derivative terms in Z reappear as an effective background-dependent mass term. Two models can therefore agree on free local kinematics and disagree on finite-wavelength dynamics.

The Maxwell action provides a useful second example. Write its real field strength as $\mathcal{F}_{\mu\nu}$, to distinguish it from the complex scalar F. Then

$$S_{\mathrm{M}} = -\frac{1}{4}\Re \int \mathrm{d}^4x\sqrt{-\gamma}\,\gamma^{\mu\alpha}\gamma^{\nu\beta}\mathcal{F}_{\mu\nu}\mathcal{F}_{\alpha\beta} = -\frac{1}{4}\int \mathrm{d}^4x\sqrt{-g}\,\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}. \tag{6.10}$$

The factors cancel as $F^2F^{-1}F^{-1} = 1$. This is the familiar four-dimensional conformal property of two-form kinetics. It does not prevent us from adding $Z(\phi)\mathcal{F}^2$, but such a factor is an additional coupling rather than an unavoidable consequence of complexification.

Exercise 5. The operator after rescaling Starting from Eq. (6.9), set $s = \frac{1}{2}\ln Z$ and $\varphi = e^{-s}\psi$. Show that the equation for ψ is

$$\left[\Box_g - m^2 - \Box_g s - (\nabla s)^2\right]\psi = 0.$$

Which term determines the wavefront cone? Which terms can scatter or amplify a finite-frequency packet?

7 The geometric scalar from a restricted curvature action

7.1 The first major dynamical choice

P1 has reduced the complex metric to $\gamma = Fg$. P2 has supplied compatible free mechanics. Neither chooses a gravitational action. A particularly economical realization is

$$S_{\mathrm{geom}} = \frac{M_{\mathrm{P}}^2}{2}\Re \int \mathrm{d}^4x\sqrt{-\gamma}\,R[\gamma], \qquad \gamma = (1+\mathrm{i}\chi)g. \tag{7.1}$$

The independent variables are the real $g_{\mu\nu}$ and χ. Allowed variations satisfy $\delta\gamma = F\,\delta g + \mathrm{i}g\,\delta\chi$. Thus Eq. (7.1) is a restricted variational problem. It is not shorthand for varying all complex tensor components independently.

7.2 Doing the conformal calculation

For $\gamma = Fg$ in four dimensions, the curvature identity is

$$R[\gamma] = F^{-1}\left[R[g] - 3\Box_g \ln F - \frac{3}{2}(\nabla \ln F)^2\right]. \tag{7.2}$$

Multiplying by the volume element and using $\Box \ln F = F^{-1}\Box F - F^{-2}(\nabla F)^2$ gives

$$\sqrt{-\gamma}R[\gamma] = \sqrt{-g}\left[FR[g] - 3\Box_g F + \frac{3}{2F}(\nabla F)^2\right]. \tag{7.3}$$

Treat boundary terms with compatible boundary conditions. Since $\nabla F = \mathrm{i}\nabla\chi$,

$$\Re\left[\frac{(\nabla F)^2}{F}\right] = -\frac{(\nabla\chi)^2}{1+\chi^2}. \tag{7.4}$$

The gravitational action becomes

$$S_{\mathrm{geom}} = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\mathrm{P}}^2}{2}R - \frac{3M_{\mathrm{P}}^2}{4(1+\chi^2)}(\nabla\chi)^2\right]. \tag{7.5}$$

The coefficient is regular and positive in the canonical kinetic sense for every real χ.

Introduce

$$f = \sqrt{\frac{3}{2}} M_{\rm P}, \qquad \phi = f \ \mathrm{arsinh}\, \chi, \qquad \mathrm{d}\phi = \frac{f\, \mathrm{d}\chi}{\sqrt{1+\chi^2}}. \tag{7.6}$$

Then

$$\boxed{S_{\rm geom} = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\rm P}^2}{2}R - \frac{1}{2}(\nabla\phi)^2\right].} \tag{7.7}$$

This equality is exact for the chosen restricted action. It is not a small-χ approximation.

7.3 A nonlinear coordinate can conceal a simple kinetic term

The relation $\chi = \sinh(\phi/f)$ is one-to-one on the real line. Near the origin $\chi \simeq \phi/f$. At large $|\chi|$, $|\phi| \simeq f \ln(2|\chi|)$. The nonlinearity in Eq. (7.5) was partly a coordinate effect on a one-dimensional field space. Every positive one-dimensional metric can locally be flattened this way.

That last sentence contains a useful warning for later. Once other response coordinates are coupled to ϕ, the *full* field space can have nonzero curvature. Flattening one coordinate's kinetic term does not flatten a multidimensional metric.

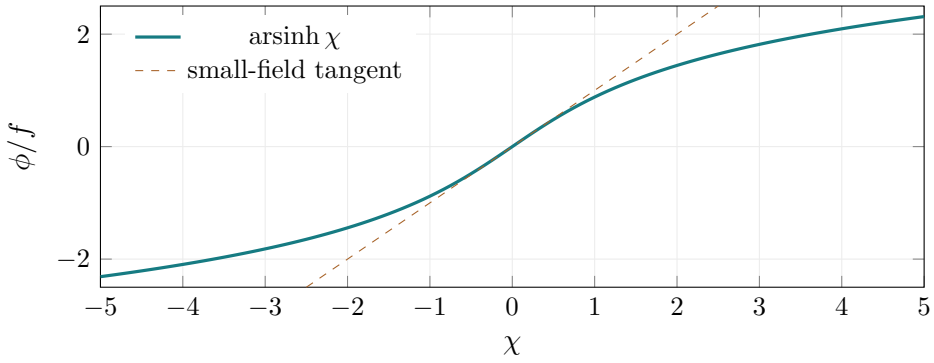


Figure 2: The exact canonical coordinate and its small-field approximation. The logarithmic growth at large $|\chi|$ is a field reparameterization, not a saturation of physical proper time.

7.4 Equations, degrees of freedom, and the GR limit

Without matter or a scalar potential, variation gives

$$M_{\rm P}^2 G_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}(\nabla\phi)^2, \qquad \Box_g\phi = 0. \tag{7.8}$$

For constant ϕ, the stress tensor vanishes and any vacuum GR solution is recovered locally. A nonconstant ϕ carries stress and can backreact even though the reference test-particle action has no direct χ force.

The regular gravitational core has three local propagating configuration degrees of freedom: two tensor polarizations and one scalar. In an ADM count, the six spatial metric components and one scalar give fourteen phase-space variables. Four first-class diffeomorphism constraints remove eight, leaving six phase-space dimensions, or three canonical pairs. Lapse and shift are multipliers in this count.

For any real null k^μ,

$$T_{\mu\nu}k^\mu k^\nu = (k^\mu\partial_\mu\phi)^2 \geq 0. \tag{7.9}$$

The minimal scalar therefore obeys the null energy condition. A spatially flat nonsingular cosmological bounce will not emerge from this positive-energy core by changing the notation from ϕ to χ.

> **Graduate extension.** Varying the unrestricted real part of a complex Einstein action near a real flat background can give opposite quadratic kinetic signs for real and imaginary spin-two perturbations. The restricted model does not retain those independent imaginary tensor perturbations. This motivates studying the constrained configuration space, but it does not prove that the selected action is unique or that every nonlinear coupled extension is stable.

Exercise 6. Canonical normalization and dimensions Derive Eq. (7.6) directly from Eq. (7.5). In four dimensions and natural units, check the mass dimensions of $\chi, \phi, f, \partial_\mu$, and the Lagrangian density. Why is ϕ/f, rather than ϕ, the correct argument of a hyperbolic function?

Part III

The response sectors

The parameter c changes a kinetic matrix, not the speed of light. At $c = 1$, the calculation changes character.

8 A two-component response and all real values of c

8.1 Choosing a constitutive relation

We now add a complex response coordinate $\Psi = u + \mathrm{i}v$, interpreted as two real scalar fields. On the shared background $F = 1 + \mathrm{i}\chi$, $\rho = \sqrt{1+\chi^2}$, choose

$$\mathcal{L}_c = -\frac{1}{2}\left[c\rho\,\partial_\mu\bar{\Psi}\,\partial^\mu\Psi + \Re\left(F\,\partial_\mu\Psi\,\partial^\mu\Psi\right)\right]. \tag{8.1}$$

The first term is of Hermitian type and the second is complex bilinear. Their relative coefficient c is real and dimensionless. This is a specific dynamical model compatible with the earlier geometry; P1 and P2 have not uniquely selected it.

Expanding is worth doing rather than trusting the notation:

$$\partial\bar{\Psi}\cdot\partial\Psi = (\partial u)^2 + (\partial v)^2, \tag{8.2}$$

$$\Re\big[F(\partial\Psi)^2\big] = (\partial u)^2 - (\partial v)^2 - 2\chi\,\partial u\cdot\partial v. \tag{8.3}$$

With $y = (u, v)^T$ this yields

$$\mathcal{L}_c = -\frac{1}{2}(G_c)_{ij}\partial_\mu y^i\partial^\mu y^j, \qquad G_c = \begin{pmatrix} c\rho+1 & -\chi \\ -\chi & c\rho-1 \end{pmatrix}. \tag{8.4}$$

The response coordinate is not a new spacetime coordinate. It is a field value at each spacetime event, much as temperature is a field without being a fifth direction in the laboratory.

8.2 The spectrum without solving a quadratic twice

Write $G_c = c\rho I + K$, where

$$K = \begin{pmatrix} 1 & -\chi \\ -\chi & -1 \end{pmatrix}.$$

Direct multiplication gives $K^2 = (1+\chi^2)I = \rho^2 I$. Since K is real symmetric and traceless, its eigenvalues are $+\rho$ and $-\rho$. Therefore

$$\boxed{\lambda_\pm = (c \pm 1)\rho, \qquad \det G_c = (c^2-1)\rho^2, \qquad \operatorname{Tr} G_c = 2c\rho.} \tag{8.5}$$

The same eigendirections work for every c; changing c changes their weights. The dependence on χ never changes the signs, because $\rho > 0$.

Range	Eigenvalue signs	Consequence in the chosen action
$c > 1$	$+, +$	Positive normal response; two regular scalar kinetic directions.
$c = 1$	$+, 0$	Rank-one positive critical response; constraints must be analyzed.
$-1 < c < 1$	$+, -$	One negative kinetic direction relative to positive-energy matter.
$c = -1$	$0, -$	A negative-sign degenerate sector, distinct from the positive critical one.
$c < -1$	$-, -$	Two negative kinetic directions with the fixed overall action sign.

8.3 Why a negative eigenvalue is a physical problem here

In a local inertial frame, for an invertible constant background matrix,

$$\pi = G_c \dot{y}, \qquad \mathcal{H}_c = \frac{1}{2}\pi^T G_c^{-1}\pi + \frac{1}{2}\sum_{\ell=1}^{3}(\partial_\ell y)^T G_c(\partial_\ell y). \tag{8.6}$$

If G_c has a negative eigenvalue, exciting its eigenmode makes this Hamiltonian arbitrarily negative in the quadratic model. Coupling that mode to positive-energy modes allows serious instabilities.

This is not the harmless minus sign in the spacetime signature. The $(-+++)$ metric was already used to turn a healthy $-\frac{1}{2}(\partial q)^2$ into $+\frac{1}{2}\dot{q}^2 - \frac{1}{2}|\nabla q|^2$. A negative *target* eigenvalue reverses that healthy kinetic sign. The two minus signs occur in different spaces.

An overall sign of an isolated action does not change its classical Euler–Lagrange equations. Once the sector is coupled to gravity and other positive-energy sectors, the relative sign matters for stress and stability. A nonsingular real field redefinition cannot alter the number of positive and negative eigenvalues of a quadratic form.

8.4 The special integers and what they do not prove

The spectrum allows every real c. The values $c = 2, 3, 4$ are selected normal sectors in the research models; $c = 1$ is the critical boundary. At $\chi = 0$ their response weights are

$$G_1 = \mathrm{diag}(2, 0), \quad G_2 = \mathrm{diag}(3, 1), \quad G_3 = \mathrm{diag}(4, 2), \quad G_4 = \mathrm{diag}(5, 3). \tag{8.7}$$

For a complex rank-n carrier, the symmetric and antisymmetric tensor squares have dimensions $n(n+1)/2$ and $n(n-1)/2$. After multiplying by $2/n$, these give weights $n+1, n-1$. This motivates an integer construction, but does not derive the full χ-dependent action or select a unique microscopic carrier.

In particular, $c = 0$ has one positive and one negative response mode; it is not pure gravity. Pure gravity is obtained by removing response fields from the action. Likewise, the labels 2 and 3 do not by themselves produce weak isospin and color, and 4 is not a proof of a new $SU(4)$ force. An integer is a useful label, but it cannot file the representation theory paperwork on our behalf.

Exercise 7. A spectrum check For $\chi = \sqrt{3}$, calculate the eigenvalues and determinant at $c = 0, 1, 2, 3, 4$. Find a normalized eigenvector of K with eigenvalue $+\rho$. Which answers change if χ is replaced by $-\chi$?

9 Canonical fields and the shared scalar interaction

9.1 Normalize at a physical reference point

For any $c > 1$, set

$$U = \sqrt{c+1}\,u, \qquad V = \sqrt{c-1}\,v, \qquad x = \phi/f. \tag{9.1}$$

Since $\rho = \cosh x$ and $\chi = \sinh x$, the normalized matrix is

$$\mathsf{G}_c = \begin{pmatrix} \dfrac{c\cosh x + 1}{c+1} & -\dfrac{\sinh x}{\sqrt{c^2-1}} \\ -\dfrac{\sinh x}{\sqrt{c^2-1}} & \dfrac{c\cosh x - 1}{c-1} \end{pmatrix}, \qquad \mathsf{G}_c(0) = I. \tag{9.2}$$

The determinant is $\cosh^2 x$. We have normalized the kinetic energy at one background, not made it constant everywhere.

Expanding around $\phi = 0$ gives

$$\mathsf{G}_c = I - \lambda_c \phi\, \sigma_1 + \frac{\phi^2}{2f^2}\,\mathrm{diag}\left(\frac{c}{c+1}, \frac{c}{c-1}\right) + O(\phi^3/f^3), \qquad \lambda_c = \frac{1}{f\sqrt{c^2-1}}. \tag{9.3}$$

The leading interaction is consequently

$$\mathcal{L}_{\phi UV} = \lambda_c \phi\, \partial_\mu U \partial^\mu V. \tag{9.4}$$

For the three selected normal sectors,

$$\lambda_2 : \lambda_3 : \lambda_4 = \frac{1}{\sqrt{3}} : \frac{1}{\sqrt{8}} : \frac{1}{\sqrt{15}}. \tag{9.5}$$

This is a ratio of canonically normalized scalar derivative couplings. It is not a prediction of the gauge-coupling ratio g_2/g_3.

9.2 Why diagonalizing the matrix is not the end

At a fixed ϕ, a rotation diagonalizes G_c. A position-dependent rotation $y = R(\phi)\widetilde{y}$ gives

$$\partial_\mu y = R\,\partial_\mu \widetilde{y} + R'\widetilde{y}\,\partial_\mu \phi.$$

The second term produces new couplings. One can simplify the matrix at a point, or even diagonalize it point by point, without eliminating all interactions. Curvature in the next section will provide a coordinate-independent test of this obstruction.

9.3 A coupling becomes an amplitude

Give ϕ, U, V masses M, m_U, m_V at the reference vacuum. For the kinematically allowed decay $\phi \to UV$, with distinguishable daughters, Eq. (9.4) gives

$$|\mathcal{M}| = \frac{|\lambda_c|}{2}|M^2 - m_U^2 - m_V^2|. \tag{9.6}$$

To see the mass combination, take the two outgoing momenta p, k. With our signature, $(p+k)^2 = -M^2$, $p^2 = -m_U^2$, and $k^2 = -m_V^2$. Hence $2p \cdot k = -(M^2 - m_U^2 - m_V^2)$. Overall vertex signs do not affect the isolated decay width.

The standard two-body phase-space integral then yields

$$\Gamma_{\phi\to UV} = \frac{\lambda_c^2(M^2 - m_U^2 - m_V^2)^2}{64\pi M}\sqrt{1 - \frac{(m_U + m_V)^2}{M^2}}\sqrt{1 - \frac{(m_U - m_V)^2}{M^2}}. \tag{9.7}$$

If all relevant daughter masses are negligible and the channels have no extra multiplicities or competing operators, their widths have ratio $1/3 : 1/8 : 1/15$. These assumptions are part of the prediction. Unequal thresholds or radiative corrections can modify the simple ratio.

Worked example 9.1 (Integrating out one heavy mediator)**.** Let $J = \sum_a \lambda_{c_a} \partial U_a \cdot \partial V_a$ and keep $-\frac{1}{2}M^2\phi^2 + \phi J$ at energies much smaller than M. The leading solution is $\phi = J/M^2$. Substitution gives

$$\Delta\mathcal{L}_{\text{eff}} = \frac{J^2}{2M^2} + \text{higher-derivative corrections.} \tag{9.8}$$

The cross-sector coefficient is $C_{ab} = \lambda_{c_a}\lambda_{c_b}/M^2$. It factorizes and has rank one. That rank-one property is generic to a single mediator; the particular ratios in Eq. (9.5) belong to the chosen CR response family.

Exercise 8. Where did the cubic interaction go? To first order in λ, make the field redefinition

$$\phi = \varphi - \frac{\lambda}{2}uv, \quad U = u + \frac{\lambda}{2}\varphi v, \quad V = v + \frac{\lambda}{2}\varphi u.$$

Verify that it cancels the cubic derivative interaction in locally flat target coordinates. Then show that a diagonal quadratic mass potential produces $\mathcal{L}_3 = \frac{\lambda}{2}(M^2 - m_U^2 - m_V^2)\varphi uv$. Why is Eq. (9.6) unchanged?

10 Target-space curvature of normal response

10.1 Two geometries at once

The theory contains a Lorentzian geometry of spacetime and a Riemannian geometry of normal field values. For one $c > 1$ response pair,

$$\mathrm{d}s_{\mathcal{T}}^2 = \mathrm{d}\phi^2 + (G_c)_{ij}(\phi)\mathrm{d}y^i\mathrm{d}y^j, \qquad q^A = (\phi, u, v). \tag{10.1}$$

The associated action is a nonlinear sigma model,

$$S_{\text{kin}} = -\frac{1}{2}\int \mathrm{d}^4x\sqrt{-g}\, h_{AB}(q)\partial_\mu q^A \partial^\mu q^B. \tag{10.2}$$

The metric $g_{\mu\nu}$ contracts spacetime indices; h_{AB} contracts field indices. The target has three dimensions, but physical spacetime still has four. Three selected response pairs sharing one ϕ will make a seven-dimensional target, not a seven-dimensional universe.

Under a smooth change of field coordinates, the coefficients h_{AB} change just as ordinary metric components do. A nonzero Riemann tensor cannot be removed by that change. Thus curvature measures interactions that survive all ordinary local field reparameterizations.

10.2 A useful curvature formula, derived

Consider the more general metric

$$\mathrm{d}s^2 = \mathrm{d}\phi^2 + H_{ab}(\phi)\mathrm{d}y^a\mathrm{d}y^b, \qquad A = H^{-1}H', \tag{10.3}$$

where primes mean ϕ derivatives. Because H has no y dependence, the only nonzero Christoffel symbols, apart from lower-index symmetry, are

$$\Gamma^{\phi}{}_{ab} = -\frac{1}{2}H'_{ab}, \qquad \Gamma^{a}{}_{\phi b} = \frac{1}{2}A^{a}{}_{b}. \tag{10.4}$$

Using the curvature convention stated in the preface, contraction gives

$$R_{\phi\phi} = -\frac{1}{2}\operatorname{Tr}A' - \frac{1}{4}\operatorname{Tr}A^2, \tag{10.5}$$

$$H^{ab}R_{ab} = -\frac{1}{2}\operatorname{Tr}A' - \frac{1}{4}(\operatorname{Tr}A)^2. \tag{10.6}$$

Adding the two traces,

$$R_{\mathcal{T}} = -\operatorname{Tr}A' - \frac{1}{4}\operatorname{Tr}A^2 - \frac{1}{4}(\operatorname{Tr}A)^2. \tag{10.7}$$

Since $A' = H^{-1}H'' - A^2$, an equivalent formula is

$$\boxed{R_{\mathcal{T}} = -\operatorname{Tr}(H^{-1}H'') + \frac{3}{4}\operatorname{Tr}(A^2) - \frac{1}{4}(\operatorname{Tr}A)^2.} \tag{10.8}$$

As a quick sign check, choose $H = a(\phi)^2 I_m$. Then $R_{\mathcal{T}} = -2m\,a''/a - m(m-1)(a'/a)^2$, the standard scalar curvature of a flat-fiber warped metric with a Riemannian base. This check is useful: the wrong factor of two in A would change both terms.

10.3 The exact single-sector answer

For $H = G_c$, the determinant immediately gives

$$\operatorname{Tr}A = \frac{\mathrm{d}}{\mathrm{d}\phi}\ln\det G_c = \frac{2}{f}\tanh x.$$

Direct multiplication supplies the remaining two quantities:

$$\operatorname{Tr}(A^2) = \frac{2}{f^2}\left[\tanh^2 x + \frac{\operatorname{sech}^2 x}{c^2-1}\right], \tag{10.9}$$

$$\operatorname{Tr}(G_c^{-1}G_c'') = \frac{2}{f^2}\left[1 + \frac{\operatorname{sech}^2 x}{c^2-1}\right]. \tag{10.10}$$

Substitution into Eq. (10.8) yields

$$\boxed{R_c = -\frac{3}{2f^2} - \frac{c^2}{2(c^2-1)f^2}\operatorname{sech}^2(\phi/f), \qquad c > 1.} \tag{10.11}$$

Every normal target in this family is curved and has negative scalar curvature. At large $|\phi|$, $f^2 R_c \to -3/2$. At the reference vacuum,

$$f^2 R_2(0) = -\frac{13}{6}, \qquad f^2 R_3(0) = -\frac{33}{16}, \qquad f^2 R_4(0) = -\frac{61}{30}.$$

For $c = 4$, using $f^2 = 3M_{\mathrm{P}}^2/2$ gives $R_4 = -M_{\mathrm{P}}^{-2}[1 + (16/45)\operatorname{sech}^2 x]$.

The sign of scalar curvature is not the sign of kinetic energy. Here G_c is positive definite, while $R_c < 0$ describes how that positive metric varies across field space. Hyperbolic geometry is not a ghost.

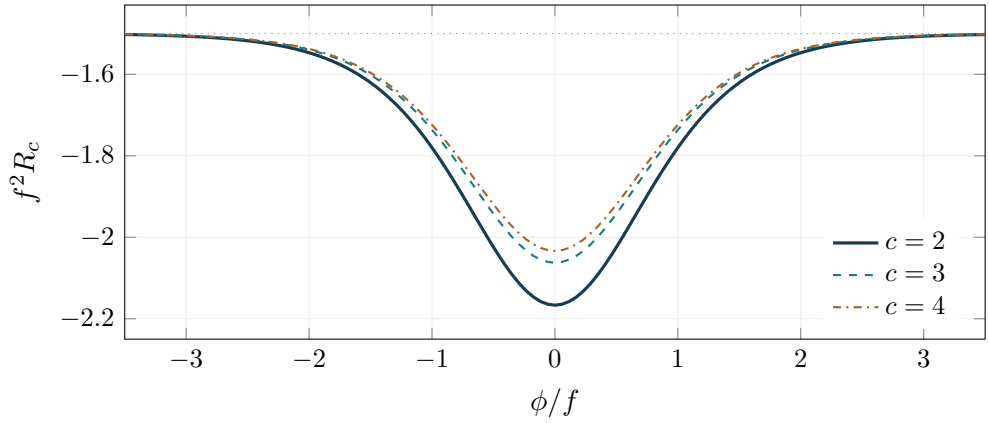


Figure 3: Exact scalar curvature of the three-dimensional normal target. Each curve approaches $-3/2$ in units of f^{-2}. Different kinetic weights produce different curvature near the common reference vacuum.

10.4 Several sectors share one base coordinate

Now take k response pairs with parameters $c_a > 1$ and the metric

$$\mathrm{d}s^2_{\mathrm{all}} = \mathrm{d}\phi^2 + \sum_{a=1}^{k} \mathrm{d}y_a^T G_{c_a}(\phi)\mathrm{d}y_a. \tag{10.12}$$

It is not a product of k independent three-dimensional targets: such a product would contain k different copies of ϕ. Here there is one shared base coordinate. The traces $\mathrm{Tr}(H^{-1}H'')$ and $\mathrm{Tr}\,A^2$ add over blocks, but $(\mathrm{Tr}\,A)^2$ also contains cross terms. Therefore

$$R_{\mathrm{all}} = \sum_{a=1}^{k} R_{c_a} - \frac{k(k-1)}{f^2}\tanh^2 x. \tag{10.13}$$

An equivalent and often more convenient form is

$$R_{\mathrm{all}} = -\frac{k(2k+1)}{2f^2} + \frac{1}{f^2}\left[k(k-1) - \frac{1}{2}\sum_{a=1}^{k}\frac{c_a^2}{c_a^2-1}\right]\mathrm{sech}^2 x. \tag{10.14}$$

For $c_a = 2, 3, 4$, the seven-field target has

$$\boxed{R_{234} = -\frac{21}{2f^2} + \frac{339}{80f^2}\,\mathrm{sech}^2 x = -\frac{501 + 339\tanh^2 x}{80f^2}.} \tag{10.15}$$

It is everywhere negative and not constant. In particular, the target cannot have an isometry translating freely along ϕ: an isometry must preserve scalar curvature.

10.5 What curvature tells a scattering experiment

In Riemann normal coordinates ξ^A around a target point,

$$h_{AB}(\xi) = \delta_{AB} - \frac{1}{3}R_{ACBD}\xi^C\xi^D + O(\xi^3). \tag{10.16}$$

The connection vanishes at that point, but curvature produces quartic derivative interactions in Eq. (10.2). The amplitude scale is schematically RE^2, hence E^2/f^2 for fixed $c > 1$. Curvature is not merely an attractive way to draw a kinetic term; it organizes invariant interaction data.

For the metric in Eq. (10.3),

$$R_{\phi a\phi b} = -\frac{1}{2}H''_{ab} + \frac{1}{4}H'_{ac}H^{cd}H'_{db}, \tag{10.17}$$

$$R_{abcd} = -\frac{1}{4}(H'_{ac}H'_{bd} - H'_{ad}H'_{bc}). \tag{10.18}$$

At the origin in normalized variables, $H'_a = -\lambda_{c_a}\sigma_1$. For two different sectors,

$$R_{U_aU_bV_aV_b} = -\frac{\lambda_{c_a}\lambda_{c_b}}{4}. \tag{10.19}$$

Thus mixed curvature exists even though the response metric has no explicit off-diagonal block connecting those sectors. The common ϕ is enough.

As $c \to 1^+$, Eq. (10.11) diverges at fixed finite ϕ and λ_c grows. This warns that the regular normal-sector expansion is failing near a rank-changing boundary. It does not prove that the properly reduced $c = 1$ system has infinite curvature: the inverse matrix used to derive the formula no longer exists there.

Exercise 9. Check the shared-base term Derive Eq. (10.13) from the trace formula and show that for $c = 2, 3$, $R_{23} = -5/f^2 + [37/(48f^2)]\,\mathrm{sech}^2\, x$. Then calculate $R_{234}(0)$ by simply adding the three single-sector values. Why does this shortcut work at the origin but fail away from it?

11 The critical sector: a zero mode is not automatically a gauge mode

11.1 Coordinates adapted to rank one

At $c = 1$, G_1 has rank one for every real ϕ. Write $a = \phi/(2f)$ and introduce

$$X = \cosh a\, u - \sinh a\, v, \qquad Y = \sinh a\, u - \cosh a\, v. \tag{11.1}$$

Differentiation gives

$$w = \mathrm{d}X - \frac{Y}{2f}\mathrm{d}\phi = \cosh a\, \mathrm{d}u - \sinh a\, \mathrm{d}v, \qquad \mathrm{d}y^T G_1 \mathrm{d}y = 2w^2. \tag{11.2}$$

Including the shared scalar, the kinetic Lagrangian is

$$\mathcal{L}_{\mathrm{crit}} = -\frac{1}{2}(\partial\phi)^2 - w_\mu w^\mu. \tag{11.3}$$

There is no independent ∂Y term, but Y multiplies $\partial\phi$. Whether it is a gauge variable, an auxiliary field, or part of a singular branch requires an actual constraint calculation.

11.2 The regular Hamiltonian branch

Use a flat time-space decomposition and write $a_Y = Y/(2f)$. The momenta are

$$p_X = 2(\dot{X} - a_Y\dot{\phi}), \qquad p_\phi = \dot{\phi} - a_Y p_X, \qquad p_Y = 0. \tag{11.4}$$

The last relation is a primary constraint. Before eliminating Y, the Hamiltonian density is

$$\mathcal{H} = \frac{1}{2}(p_\phi + a_Y p_X)^2 + \frac{1}{4}p_X^2 + \frac{1}{2}|\nabla\phi|^2 + |\nabla X - a_Y\nabla\phi|^2. \tag{11.5}$$

Preserving $p_Y = 0$ under time evolution gives the secondary constraint

$$C_2 = N + \frac{Y}{2f}D = 0, \qquad D = \frac{1}{2}p_X^2 + |\nabla\phi|^2, \quad N = \frac{1}{2}p_X p_\phi - \nabla X \cdot \nabla\phi. \tag{11.6}$$

The relevant Poisson bracket is

$$\{p_Y(\boldsymbol{x}), C_2(\boldsymbol{x}')\} = -\frac{D(\boldsymbol{x})}{2f}\delta(\boldsymbol{x} - \boldsymbol{x}'). \tag{11.7}$$

For $D > 0$, the constraints are *second class*: they remove a canonical pair without generating a gauge orbit. They give $Y = -2fN/D$ and leave two configuration degrees of freedom, including ϕ, from the original three.

Substitution gives

$$\mathcal{H}_{\text{red}} = \frac{1}{2}p_\phi^2 + \frac{1}{4}p_X^2 + \frac{1}{2}|\nabla\phi|^2 + |\nabla X|^2 - \frac{N^2}{D}. \tag{11.8}$$

The negative last term looks worrying until the expression is completed as a square. Let $A = (p_X/\sqrt{2}, \nabla\phi)$ and $B = (p_\phi/\sqrt{2}, -\nabla X)$. Then $D = |A|^2$, $N = A \cdot B$, and

$$\mathcal{H}_{\text{red}} = \frac{1}{2}|A|^2 + \left|B - \frac{A \cdot B}{|A|^2}A\right|^2 \geq 0. \tag{11.9}$$

This is a genuine positivity result on the regular branch. It says nothing by itself about division by D at $D = 0$, where the constraint rank changes and a separate analysis is necessary.

11.3 Contact geometry, with a physical caution

On the three-dimensional field space,

$$w \wedge \mathrm{d}w = -\frac{1}{2f}\mathrm{d}X \wedge \mathrm{d}Y \wedge \mathrm{d}\phi \neq 0. \tag{11.10}$$

Thus w is a contact form. In homogeneous reduction with $r = p_X \neq 0$, the constraint gives $Y = -2fp_\phi/r$, and the canonical one-form is $p_X\mathrm{d}X + p_\phi\mathrm{d}\phi = rw$. This is the familiar symplectization of a contact manifold, with the overall two-form sign set by convention.

Contact and symplectic structures are valuable, but they are already present in classical mechanics. They do not supply a universal quantum action unit or a Born probability rule. In particular, the two original response coordinates should not simply be declared a conjugate quantum pair.

The transformation

$$X \mapsto X + h(\phi), \qquad Y \mapsto Y + 2fh'(\phi) \tag{11.11}$$

preserves w. Here h is a function of the field value ϕ, not an arbitrary independent function of spacetime. The regular constraints remain second class. A zero eigenvalue plus a suggestive symmetry is therefore not sufficient to identify a photon or Maxwell gauge symmetry.

11.4 Propagation must be checked after reduction

For every normal sector, $P_c(k) = G_c\, g^{\mu\nu}k_\mu k_\nu$. Since G_c is invertible, its characteristic cone is the real null cone. At $c = 1$, taking $\det P_1 = 0$ gives no information: it vanishes for every k. One must first remove constraints and identify physical propagating variables.

For example, around a background $\bar{w} = \bar{Y} = 0$ with $n_\mu = \partial_\mu\bar{\phi}$, the quadratic response term contains $-(\partial_\mu\delta X - \delta Y\, n_\mu/(2f))^2$. If $n^2 \neq 0$, algebraic elimination of δY projects the principal tensor to

$$P^{\mu\nu}_{\text{proj}} = g^{\mu\nu} - \frac{n^\mu n^\nu}{n^2}. \tag{11.12}$$

This tensor is degenerate and generally does not have the ordinary Lorentz cone. For a timelike n, it projects onto spatial directions; that particular background can also lie on a singular Hamiltonian stratum. For a spacelike n, one spatial derivative direction is removed. If n is null or zero, the displayed elimination is invalid.

This calculation identifies a problem to analyze, not a ready-made claim of superluminal signal transmission. The remaining constraints, background branch, and well-posed initial-value problem decide which disturbances are physical. Any physical signal incompatible with P1 would exclude that realization or require changing its dynamics.

Exercise 10. Second class, and positive Derive C_2 from Eq. (11.5), verify Eq. (11.7), and complete the square to obtain Eq. (11.9). State exactly where your derivation uses $D > 0$.

Part IV

Building physical models

Shared geometry correlates interactions. Particle identities, potentials, and initial conditions still have to be specified.

12 One action, one shared source

12.1 Combining the pieces without counting the scalar twice

A useful normal-sector model is

$$S = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\mathrm{P}}^2}{2}R - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}\sum_a (G_{c_a})_{ij}\nabla_\mu y_a^i \nabla^\mu y_a^j \right.$$
$$\left. - V(\phi, y_a) - \sum_b \frac{Z_b(\phi)}{4g_{b0}^2}\mathcal{F}^A_{b\mu\nu}\mathcal{F}_b^{A\mu\nu} + \mathcal{L}_{\mathrm{other}}\right]. \tag{12.1}$$

Here $c_a > 1$, $Z_b > 0$, and $\mathcal{L}_{\mathrm{other}}$ can include specified charged matter. The gauge index A is absent for an Abelian field. If the critical module is retained, add $-w_\mu w^\mu$ and analyze its constraints separately. For the normal examples in this part, we use the consistent classical truncation $X = \text{constant}$, $Y = 0$, $w = 0$, assuming no other term depends on X or Y.

There is one ϕ kinetic term, regardless of how many response sectors are added. Summing a complete gravity-plus-scalar action for each sector would silently change the normalization of ϕ and therefore change every coupling λ_c. The shared field is a physical identification, not a typographical convenience.

12.2 Derive the scalar equation from every term

For the action displayed, with $\mathcal{L}_{\mathrm{other}}$ independent of ϕ, variation gives

$$\Box_g\phi = V_{,\phi}\frac{1}{2}\sum_a (G'_{c_a})_{ij}\nabla y_a^i \cdot \nabla y_a^j + \sum_b \frac{Z_b'}{4g_{b0}^2}\mathcal{F}_b^2. \tag{12.2}$$

If the critical term is present, add $\nabla_\mu(Yw^\mu/f)$ on the right. A ϕ-dependent Yukawa coupling or additional portal contributes its own variation. There is no reason to retain only the source term one happens to like.

The response equations are

$$\nabla_\mu\left[(G_{c_a})_{ij}\nabla^\mu y_a^j\right] = V_{,y_a^i}. \tag{12.3}$$

The metric equation is $M_{\mathrm{P}}^2 G_{\mu\nu} = T_{\mu\nu}^{\mathrm{total}}$. The potential contributes $-g_{\mu\nu}V$ once. Assigning it separately to several sector stress tensors and then adding those tensors would count the same energy more than once.

12.3 Energy exchange is not energy nonconservation

For one potential-free response sector,

$$T_{\mu\nu}^{(c)} = (G_c)_{ij}\partial_\mu y^i \partial_\nu y^j - \frac{1}{2}g_{\mu\nu}(G_c)_{ij}\nabla y^i \cdot \nabla y^j. \tag{12.4}$$

Its field equation implies

$$\nabla_\mu T^{(c)\mu}{}_\nu = -\frac{1}{2}(G_c')_{ij}\nabla y^i \cdot \nabla y^j \, \partial_\nu \phi. \tag{12.5}$$

Meanwhile $\nabla_\mu T^{(\phi)\mu}{}_\nu = (\Box\phi)\partial_\nu\phi$. The matching term in Eq. (12.2) cancels the response divergence. The sectors exchange energy, while the total stress obeys covariant conservation. This cancellation is a practical sign check on the entire action.

12.4 A real GR limit requires the frozen equation too

Setting $\phi = \phi_0$ and the responses to stationary values is a candidate truncation. It is consistent only if Eq. (12.2) vanishes there for every matter configuration being retained. With $Z_b' = 0$, ϕ-independent visible couplings, and a stationary potential, this can hold. If $Z_b' \neq 0$, a nonzero gauge background can source ϕ again.

A background at the bottom of a potential is therefore not by itself a proof that every solution of GR with matter is embedded. Nor does a classical truncation remove quantum fluctuations of the omitted fields. Classical solution embedding, low-energy decoupling, and quantum equivalence are three separate calculations.

Exercise 11. Check the energy transfer For a homogeneous, potential-free response on Minkowski spacetime, $E_y = \frac{1}{2}\dot{y}^T G_c \dot{y}$. Use $\frac{\mathrm{d}}{\mathrm{d}t}(G_c\dot{y}) = 0$ to show that $\dot{E}_y = -\frac{1}{2}\dot{\phi}\,\dot{y}^T G_c' \dot{y}$. Use Eq. (12.2) to show that the homogeneous scalar energy changes by the opposite amount.

13 Internal gauge structure: what the extra assumptions buy

13.1 An internal vector space is not a tangent space

A gauge field compares internal states at neighboring spacetime points. Mathematically, one introduces a vector bundle whose fiber carries those states and a connection that transports them. The spacetime metric specifies lengths and cones; it does not, by itself, specify the dimension or representation of this internal fiber.

An explicit model in the CR sources takes a complex rank-five bundle with a positive Hermitian metric, a preserved splitting, and a trivialized total determinant:

$$E = E_3 \oplus E_2, \qquad \mathrm{rank}_{\mathbb{C}}\, E_3 = 3, \quad \mathrm{rank}_{\mathbb{C}}\, E_2 = 2, \qquad \det E \simeq \mathbf{1}. \tag{13.1}$$

Unitary transformations preserving the two subbundles are pairs (U_3, U_2), and preserving the determinant imposes $\det U_3 \det U_2 = 1$. The resulting group is

$$S(U(3)\times U(2)) \simeq \frac{SU(3)\times SU(2)\times U(1)}{\mathbb{Z}_6}. \tag{13.2}$$

The group is derived from the structures in Eq. (13.1). Those structures themselves are model inputs. This conditional derivation is useful without being a derivation of the entire Standard Model from a single complex scalar.

13.2 Seeing hypercharge in the blocks

A traceless generator is

$$Y_{\mathrm{hyp}} = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \tag{13.3}$$

With Hermitian generators and $D = \mathrm{d} - \mathrm{i}\mathcal{A}$, write

$$\mathcal{A} = \begin{pmatrix} \mathcal{C} - \frac{1}{3}\mathcal{B}I_3 & 0 \\ 0 & \mathcal{W} + \frac{1}{2}\mathcal{B}I_2 \end{pmatrix}, \qquad \operatorname{Tr}\mathcal{C} = \operatorname{Tr}\mathcal{W} = 0. \tag{13.4}$$

Here $\mathcal{C}, \mathcal{W}, \mathcal{B}$ are independent connections with curvature $\mathcal{F} = \mathrm{d}\mathcal{A} - \mathrm{i}\mathcal{A} \wedge \mathcal{A}$. The algebra has $8+3+1 = 12$ generators. With ordinary Yang–Mills kinetic terms and constraints, each unbroken massless generator has two transverse physical polarizations.

The quotient can be checked directly using the covering map

$$(g_3, g_2, e^{\mathrm{i}\alpha}) \longmapsto \operatorname{diag}(e^{-2\mathrm{i}\alpha} g_3, e^{3\mathrm{i}\alpha} g_2). \tag{13.5}$$

Its kernel has six elements. The quotient affects globally allowed representations; it does not equate the three independent gauge couplings. In the connection convention above, couplings appear in the kinetic coefficients, not a second time inside $\mathcal{A}$.

13.3 A small representation-theory laboratory

Let E_3 transform as $(\mathbf{3}, \mathbf{1})_{-1/3}$ and E_2 as $(\mathbf{1}, \mathbf{2})_{1/2}$. Then

$$\Lambda^2 E = \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2 = (\bar{\mathbf{3}}, \mathbf{1})_{-2/3} \oplus (\mathbf{3}, \mathbf{2})_{1/6} \oplus (\mathbf{1}, \mathbf{1})_1. \tag{13.6}$$

Together with $E^* = (\bar{\mathbf{3}}, \mathbf{1})_{1/3} \oplus (\mathbf{1}, \mathbf{2})_{-1/2}$, these are the familiar left-handed representation assignments of one Standard Model generation. The dimensions add to $10 + 5 = 15$. A sterile neutrino can be added as another singlet.

This elegant bookkeeping still does not tell us why matter is described by Weyl spinors, why there are three generations, what their masses are, or what selects their statistics. Lorentz spin belongs to a spinor bundle; internal multiplicity belongs to a separate factor. A three-component spatial vector is not a color triplet merely because both have three entries.

> **Graduate extension.** For a representation decomposed as $\mathcal{V} = \bigoplus_\alpha R_\alpha \otimes M_\alpha$, with inequivalent irreducible Lorentz representations R_α, the Lorentz-commuting internal algebra acts on the multiplicity spaces: $\operatorname{End}_{\mathrm{Lorentz}}(\mathcal{V}) = \bigoplus_\alpha \operatorname{End}(M_\alpha)$ under the usual complex irreducibility assumptions. Three copies of the same Lorentz representation can carry color. The three components inside one irreducible Lorentz representation do not supply the same internal freedom.

13.4 Why the response fields were kept as singlets

The bilinear $(\partial\Psi)^2$ in Eq. (8.1) is not invariant under an arbitrary complex charged representation. Promoting Ψ to a charged multiplet therefore requires an appropriate invariant bilinear, a covariant derivative, and all of the representation components. In the minimal presentation, u_c, v_c remain gauge singlets while charged matter is introduced separately.

The exact Maxwell cancellation in Eq. (6.10) suggests the baseline $Z_b = 1$. A choice $Z_b(\phi) = \cosh(\phi/f)$ is allowed and positive but is an extra interaction. At a slowly varying fixed background its canonically normalized coupling is $g_b(\phi) = g_{b0}/\sqrt{Z_b(\phi)}$. Measurements of varying couplings would test this chosen function and the background history, not complex geometry in isolation.

Exercise 12. Charges by addition Recover the three hypercharges in $\Lambda^2 E$ by adding the charges of the factors. Then use the five left-handed multiplets just listed to check $\sum d_i Y_i = 0$ and $\sum d_i Y_i^3 = 0$, where d_i includes color and weak multiplicities. These are two anomaly checks, not a substitute for checking the mixed non-Abelian anomalies and global consistency.

14 Cosmology: what the simplest scalar actually does

14.1 Expansion as an energy diagnostic

In a spatially flat homogeneous universe,

$$\mathrm{d}s^2 = -\mathrm{d}\tau^2 + a(\tau)^2\mathrm{d}\boldsymbol{x}^2, \qquad H = \frac{\dot{a}}{a}. \tag{14.1}$$

For a canonical homogeneous scalar with potential V,

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V, \qquad p_\phi = \frac{1}{2}\dot{\phi}^2 - V, \tag{14.2}$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0, \qquad 3M_{\mathrm{P}}^2H^2 = \rho_{\mathrm{total}}. \tag{14.3}$$

Dots now denote cosmic proper time. The term $3H\dot{\phi}$ describes the redshifting of homogeneous scalar motion during expansion. For a separately conserved component, $\dot{\rho} + 3H(\rho + p) = 0$. If $w = p/\rho$ is constant,

$$\rho \propto a^{-3(1+w)}. \tag{14.4}$$

14.2 The free geometric scalar is stiff matter

Set $V = 0$. Then $a^3\dot{\phi}$ is constant, and

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 \propto a^{-6}, \qquad p_\phi = \rho_\phi, \qquad w = 1. \tag{14.5}$$

If this component dominates, the expanding solution is

$$a \propto \tau^{1/3}, \qquad \phi = \phi_0 \pm \sqrt{\frac{2}{3}}M_{\mathrm{P}}\ln(\tau/\tau_0). \tag{14.6}$$

In the original geometric variable, $\chi = \sinh[\phi_0/f \pm (2/3)\ln(\tau/\tau_0)]$.

The word *scalar* does not determine an equation of state. A homogeneous kinetic scalar gives $w = 1$; an isotropic gas of relativistic scalar quanta gives $w \simeq 1/3$; a rapidly oscillating massive condensate can give $w \simeq 0$; a slowly varying potential-dominated field can give $w \simeq -1$. The field type is the same, but the state and dynamics differ.

Component or regime	w	Density scaling
Homogeneous free scalar	1	a^{-6}
Relativistic particle gas	1/3	a^{-4}
Cold matter or quadratic oscillations	$\simeq 0$	$\simeq a^{-3}$
Constant potential	-1	Constant

14.3 How a massive response can behave as cold matter

On the reference branch $\phi = 0$, take a canonically normalized field S with potential $m_S^2S^2/2$. Provided the other source equations allow this branch,

$$\ddot{S} + 3H\dot{S} + m_S^2S = 0. \tag{14.7}$$

When $m_S \gg H$ and the background changes little in one oscillation, write $S = A(\tau)\cos(m_S\tau + \delta)$. Averaging the equation over an oscillation gives $A \propto a^{-3/2}$. The average kinetic and potential energies agree, so

$$\langle p_S\rangle \simeq 0, \qquad \langle\rho_S\rangle \simeq \frac{1}{2}m_S^2A^2 \propto a^{-3}. \tag{14.8}$$

The amplitude remains initial-state data. This calculation supplies a cold-matter-like branch, not a numerical prediction of the observed dark-matter abundance.

14.4 Potentials, bounces, and stability

A constant V_0 acts as $\Lambda = V_0/M_{\rm P}^2$ and should be counted once. A slowly rolling potential requires the usual comparison of slope, curvature, and Hubble friction. Neither is fixed by P1/P2.

For flat Einstein cosmology,

$$\dot{H} = -\frac{\rho + p}{2M_{\rm P}^2}. \tag{14.9}$$

At a nonsingular bounce, H must cross from negative to positive, so $\dot{H} > 0$ is needed near the crossing. Positive canonical scalar and normal-response kinetics satisfy the null energy condition and do not provide this in a spatially flat model. Spatial curvature, modified gravity, or additional dynamics can change the analysis, but must be specified.

A homogeneous solution is also not a perturbative stability proof. One must expand all coupled fields, including $\delta\phi$, response mixing, and metric perturbations. An oscillatory background can drive parametric resonance. The sign of an instantaneous mass alone does not replace a coupled mode or Floquet analysis.

Exercise 13. Derive the stiff solution Combine $a^3\dot{\phi} =$ constant with the scalar-dominated Friedmann equation to derive Eq. (14.6). Then show that a constant rescaling of the action leaves the scalar equation unchanged but changes its gravitational normalization unless all corresponding couplings are transformed consistently.

15 A dark-sector example, including the decay nobody invited

15.1 The minimal mass model

For $c = 4$, choose the potential

$$V = \frac{1}{2}M_\phi^2\phi^2 + \frac{5}{2}m_4^2u_4^2, \qquad S = \sqrt{5}\,u_4, \quad V_4 = \sqrt{3}\,v_4. \tag{15.1}$$

At the origin the masses are M_ϕ, m_4, and zero. The massless V_4 mode survives because the potential is independent of v_4. A zero background value does not remove its fluctuations.

On a nearly constant geometric background, physical masses solve the *generalized* eigenvalue problem $\det(K_{\rm mass} - m^2_{\rm phys}G_4) = 0$, with $K_{\rm mass} = {\rm diag}(5m_4^2, 0)$. Expanding the determinant gives one zero root and

$$m_S^2(\phi) = m_4^2\frac{4\cosh(\phi/f) - 1}{3\cosh^2(\phi/f)}. \tag{15.2}$$

Dividing $5m_4^2$ by the first diagonal entry of G_4 would miss the off-diagonal kinetic mixing. Generalized eigenvalues are the appropriate tool whenever the kinetic metric is not the identity.

Near the origin, $m_S^2 = m_4^2[1 - \phi^2/(3f^2) + O(\phi^4/f^4)]$. There is no linear diagonal scalar charge there. Nevertheless the off-diagonal vertex $\lambda_4\phi\,\partial S\cdot\partial V_4$ remains.

15.2 Symmetry protection is a statement about the full spectrum

The isolated three-field model admits

$$P_\phi: \quad (\phi, u_4, v_4) \mapsto (-\phi, u_4, -v_4), \tag{15.3}$$

$$P_{y_4}: \quad (\phi, u_4, v_4) \mapsto (\phi, -u_4, -v_4). \tag{15.4}$$

Under their product, S and ϕ are odd while V_4 is even. If every interaction and the vacuum preserve this parity, and $m_4 < M_\phi$, then S is the lightest odd particle of the isolated model. A single S cannot decay into only even states.

If $m_4 > M_\phi$, the process $S \to \phi V_4$ is open, with tree-level width

$$\Gamma_{S\to\phi V_4} = \frac{m_4^3}{960\pi f^2}\left(1 - \frac{M_\phi^2}{m_4^2}\right)^3 . \tag{15.5}$$

The parity is conserved: both the initial state and the final ϕ carry odd parity. A discrete symmetry forbids the wrong parity, not every decay one would prefer to avoid.

15.3 Adding the other sectors changes the conclusion

To preserve the shared kinetic interactions, extend P_ϕ by $v_n \mapsto -v_n$, $u_n \mapsto u_n$ for the other responses. Under the combined dark parity, V_2, V_3 are then odd. If one of them is lighter than S, the isolated stability argument no longer applies.

Even for $M_\phi > m_4$, the heavy-mediator interaction Eq. (9.8) contains

$$\frac{\lambda_4\lambda_n}{M_\phi^2}(\partial S \cdot \partial V_4)(\partial U_n \cdot \partial V_n). \tag{15.6}$$

It can permit $S \to V_4 + U_n + V_n$ through a virtual ϕ if the mass threshold is open. The decay conserves the parity. Closing a two-body channel does not close every channel in the larger theory.

For a genuinely stable eigenstate, compare its mass with *all* lighter odd physical states and verify the symmetry of the complete action and vacuum. If a channel is open but very slow, the correct description is long-lived or decaying dark matter, with a calculable lifetime. The distinction is physical, not semantic.

Exercise 14. A generalized mass eigenvalue Compute $\det[\mathrm{diag}(5m_4^2, 0) - \mu G_4]$ and recover Eq. (15.2). Check its value and first derivative at $\phi = 0$. Then explain why those checks alone cannot establish the stability of the massive particle.

Part V

Quantum structure and physical observation

Complex geometry supplies useful structures. To obtain quantum physics, one must also identify states, observables, and the operations that connect them.

16 How much quantum mechanics is already in the geometry?

16.1 Pauli matrices appear, but the interpretation still matters

The normalized response involution is

$$J_{\mathrm{CR}} = \frac{1}{\rho}\begin{pmatrix} 1 & -\chi \\ -\chi & -1 \end{pmatrix} = \frac{\sigma_3 - \chi\sigma_1}{\sqrt{1+\chi^2}}, \qquad J_{\mathrm{CR}}^2 = I. \tag{16.1}$$

Set $\theta = \arctan\chi$. Then $J_{\mathrm{CR}} = \cos\theta\,\sigma_3 - \sin\theta\,\sigma_1$. Its positive projector is $P_+(\theta) = (I + J_{\mathrm{CR}})/2$. The Pauli identity $\mathrm{Tr}[(\boldsymbol{a}\cdot\boldsymbol{\sigma})(\boldsymbol{b}\cdot\boldsymbol{\sigma})] = 2\boldsymbol{a}\cdot\boldsymbol{b}$ gives

$$\mathrm{Tr}[P_+(\theta)P_+(\theta')] = \frac{1+\cos(\theta-\theta')}{2} = \cos^2\frac{\theta-\theta'}{2}. \tag{16.2}$$

This is exactly the mathematical shape of a spin-1/2 overlap probability. It is a useful structural connection. To interpret it as a physical probability, however, we need a preparation rule, a measurement rule, and a justification for why the trace is the correct statistical weight.

The family also traces only a one-parameter arc of projectors: $\theta \in (-\pi/2, \pi/2)$. It does not span the full Bloch sphere. For $c > 1$, normalizing the positive response matrix itself gives

$$\frac{G_c}{\mathrm{Tr}\,G_c} = \frac{1}{2}\left(I + \frac{1}{c}J_{\mathrm{CR}}\right), \qquad \mathrm{Tr}\left[\left(\frac{G_c}{\mathrm{Tr}\,G_c}\right)^2\right] = \frac{1}{2}(1+c^{-2}). \tag{16.3}$$

This matrix has the algebraic form of a mixed qubit state with Bloch radius $1/c$. A positive matrix is an excellent candidate for many roles; an experiment must determine which role it actually plays.

16.2 A classical route to a projective sphere

Two equal-frequency classical oscillators already give an instructive example. Define

$$z_j = \frac{\sqrt{\omega}\,q_j + \mathrm{i}p_j/\sqrt{\omega}}{\sqrt{2}}, \qquad I = |z_1|^2 + |z_2|^2, \qquad H = \omega I. \tag{16.4}$$

Fixing $I = I_0 > 0$ gives a three-sphere in $\mathbb{C}^2$. If common phase is identified, the quotient is $S^3/U(1) \simeq \mathbb{CP}^1$, a two-sphere. A representative is

$$z = \sqrt{I_0}\,e^{\mathrm{i}\alpha}\begin{pmatrix} \cos(\vartheta/2) \\ e^{\mathrm{i}\varphi}\sin(\vartheta/2) \end{pmatrix}. \tag{16.5}$$

The first oscillator's action fraction is $|z_1|^2/I_0 = \cos^2(\vartheta/2)$. With orientation chosen for positive area, the reduced area form has total area $2\pi I_0$:

$$\Omega_{\mathrm{red}} = \frac{I_0}{2}\sin\vartheta\,\mathrm{d}\vartheta\wedge\mathrm{d}\varphi, \qquad \int_{\mathbb{CP}^1}\Omega_{\mathrm{red}} = 2\pi I_0. \tag{16.6}$$

This positive-area convention can differ by an overall sign from the particle symplectic convention in the preface; it changes no counting.

Projective geometry and squared overlaps have appeared in a classical system. If one now imposes geometric quantization, $I_0/\hbar = k \in \mathbb{Z}_{\geq 0}$, the standard holomorphic construction gives $k+1$ states. A two-state system requires $k = 1$. The quantization prescription and the action unit are additional inputs. Likewise, quotienting common phase is physically appropriate only if the allowed probes cannot resolve it.

16.3 Three different antisymmetric structures

It is useful to distinguish:

1. a two-form on a fixed response fiber, such as $\sqrt{\det G_c}\,\mathrm{d}u\wedge\mathrm{d}v$;
2. the canonical field phase-space form involving fields and their momenta, such as $\int(\delta\pi_u\wedge\delta u + \delta\pi_v\wedge\delta v)\,\mathrm{d}^3x$;
3. a quantum commutator of operators representing actual observables.

The first can define a compatible complex structure on a two-dimensional positive fiber. The second generates classical dynamics and normally has $\{u, v\} = 0$, because both are configuration coordinates. The third requires a representation and a state space. These structures can be related by a physical reduction or quantization, but equality does not follow from the fact that all three are antisymmetric.

16.4 Positive classical energy does not fix an action quantum

An oscillator with equation $\ddot{q}+\omega^2 q = 0$ can be quantized with $[\hat{q},\hat{p}] = \mathrm{i}\kappa I$ for any chosen positive action scale κ. Its normalized ground-state correlator is

$$\langle\hat{q}(t)\hat{q}(s)\rangle = \frac{\kappa}{2\omega}e^{-\mathrm{i}\omega(t-s)}. \tag{16.7}$$

The frequency and classical trajectories do not determine κ. Similarly, multiplying a classical action by a positive constant preserves its stationary paths while rescaling canonical momenta. Operational calibration is needed to connect an action normalization to the measured Planck constant.

One can certainly use ordinary quantum theory to quantize a suitable CR effective action. That is a legitimate calculational program. Deriving quantum theory and its common action scale *from* CR would be a stronger result.

16.5 A concrete test: positivity of the quantum state

Assume for the moment that the physical observables obey $[\hat{q},\hat{p}] = \mathrm{i}\kappa I$ and define

$$W(\xi) = e^{\mathrm{i}(\xi_q\hat{q}+\xi_p\hat{p})}, \qquad \sigma(\xi,\eta) = \xi_q\eta_p - \xi_p\eta_q. \tag{16.8}$$

The Baker–Campbell–Hausdorff formula gives

$$W(\xi)W(\eta) = e^{-\mathrm{i}\kappa\sigma(\xi,\eta)/2}W(\xi+\eta). \tag{16.9}$$

For a positive state with characteristic function $\Phi(\xi) = \langle W(\xi)\rangle$, positivity of $\langle A^\dagger A\rangle$ for $A = \sum_j a_j W(\xi_j)$ requires

$$M_{jk} = e^{\mathrm{i}\kappa\sigma(\xi_j,\xi_k)/2}\Phi(\xi_k - \xi_j) \succeq 0 \quad \text{for every finite set } \{\xi_j\}. \tag{16.10}$$

This is a positivity condition on the quantum algebra, with its nontrivial phase factor.

Take the deterministic classical state at $q = p = 0$. Its ordinary characteristic function is $\Phi_{\mathrm{cl}} = 1$. At $\xi_0 = (0,0)$, $\xi_1 = (a,0)$, $\xi_2 = (0,b)$, the quantum test matrix is

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & e^{\mathrm{i}\theta} \\ 1 & e^{-\mathrm{i}\theta} & 1 \end{pmatrix}, \qquad \theta = \frac{\kappa ab}{2}, \qquad \det M = -4\sin^2(\theta/2). \tag{16.11}$$

For generic nonzero area it fails positivity. Ordinary positive probability on classical phase space is not sufficient to be a quantum state for a nonzero-κ Weyl algebra.

For an uncorrelated Gaussian $\Phi = e^{-(V_q\xi_q^2 + V_p\xi_p^2)/2}$, the small-$a, b$ determinant is

$$\det M = a^2b^2\left(V_qV_p - \frac{\kappa^2}{4}\right) + O(a^4b^2 + a^2b^4). \tag{16.12}$$

Thus positivity implies $V_qV_p \geq \kappa^2/4$. The uncertainty bound emerges from an algebra and a state-positivity condition. For non-Gaussian states, checking only second moments is not enough; Eq. (16.10) tests more.

Exercise 15. A three-point quantum test Calculate the determinant in Eq. (16.11) directly. Why does choosing just one special area with $\theta = 2\pi n$ not rescue the classical point state as a quantum state? Which quantifier in Eq. (16.10) prevents that shortcut?

17 Probabilities need preparation and recording dynamics

17.1 The same geometry admits different measures

Suppose an outcome occurs when a hidden preparation variable $z \in [0,1]$ is below a setting-dependent q. If z is uniform, the probability is q. If its density is $6z(1-z)$, the probability is

$$P(z \leq q) = 3q^2 - 2q^3. \tag{17.1}$$

Both densities are positive and normalized. The geometry of the interval did not choose between them.

A physically motivated uniform ratio can nevertheless arise. At fixed configuration, a normal response has kinetic energy $K = \frac{1}{2}p^T G_c^{-1} p$. In a Gibbs ensemble at inverse temperature β, whiten momentum by $\widetilde{p} = G_c^{-1/2}p$. For two quadratic momentum components, K is exponentially distributed: $p(K) = \beta e^{-\beta K}$ for $K \geq 0$. For two independent sectors at the same temperature, set $z = K_1/(K_1 + K_2)$ and $r = K_1 + K_2$. The Jacobian is r, so

$$p(z) = \int_0^\infty \beta^2 e^{-\beta r} r\, \mathrm{d}r = 1, \qquad 0 \leq z \leq 1. \tag{17.2}$$

This is a derivation with explicit ensemble assumptions. It does not require equal raw coordinate variances. It does require independence, the stated number of quadratic degrees of freedom, and a common temperature. Driving and conditioning can change the ensemble.

17.2 A complete first-record model

Let channels accumulate work

$$W_i(t) = a_i q_i \, \mathcal{A}(t), \qquad \mathcal{A}(0) = 0, \quad \mathcal{A}'(t) \geq 0, \tag{17.3}$$

where $a_i > 0$ are gains and $q_i \geq 0$ are encoded signal weights. A channel records when its work exceeds an independent random threshold K_i with survival probability $P(K_i > k) = e^{-\beta_i k}$. The probability that no channel has recorded by t is

$$P_\varnothing(t) = \prod_i e^{-\beta_i W_i(t)} = e^{-R\mathcal{A}(t)}, \qquad r_i = \beta_i a_i q_i, \quad R = \sum_i r_i. \tag{17.4}$$

To get a first record in channel i during $\mathrm{d}t$, multiply the survival of all channels by that channel's instantaneous hazard:

$$p(i,t) = r_i \mathcal{A}'(t) e^{-R\mathcal{A}(t)}. \tag{17.5}$$

Integrating to a cutoff T gives

$$P(i,\ t \leq T) = \frac{r_i}{R}[1 - e^{-R\mathcal{A}(T)}], \qquad P(i \mid \text{some record by } T) = \frac{r_i}{R}. \tag{17.6}$$

If $\sum_i q_i = 1$ and all $\beta_i a_i$ are equal, the conditional channel probabilities are q_i. Unlike a bare overlap formula, this model also predicts waiting times and missed events. With a finite work budget $\mathcal{A}(\infty) < \infty$, there can be a nonzero probability of never recording.

Worked example 17.1 (Unequal detector gains)**.** Take $q_1 = 1/4$, $q_2 = 3/4$, but $\beta_1 a_1 = 2\beta_2 a_2$. Then $P(1 \mid \text{record}) = (2/4)/(2/4 + 3/4) = 2/5$. The same encoded weights now give a different observed frequency. Equal effective hazards are a physical calibration condition, not an algebraic consequence of normalization.

Different threshold laws also matter. If $P(K_i > k) = e^{-\beta_i k^\alpha}$ with common $\alpha > 0$, the same calculation gives weights proportional to $\beta_i a_i^\alpha q_i^\alpha$. Selecting a linear probability law therefore needs a reason for the appropriate threshold statistics or an operational additivity principle.

17.3 A first click is not a full measurement theory

A first-record law does not specify the state left after a click. Repeated and sequential experiments require conditional updates, no-click evolution, and composition with other apparatus. In a quantum model an instrument would supply maps $\varrho \mapsto M_i \varrho M_i^\dagger$ and probabilities $\mathrm{Tr}(M_i \varrho M_i^\dagger)$, with appropriate completeness conditions. Writing such maps is already importing quantum state structure unless their physical derivation has been supplied.

Likewise, a stationary distribution is not automatically a martingale that produces Born-like absorption probabilities. For a diffusion $\mathrm{d}q = b(q)\mathrm{d}t + \sqrt{a(q)}\mathrm{d}W_t$ absorbed at 0 or 1, the probability $h(q)$ of reaching 1 first solves

$$b(q)h'(q) + \frac{1}{2}a(q)h''(q) = 0, \qquad h(0) = 0, \quad h(1) = 1. \tag{17.7}$$

Under regularity and almost-sure absorption, $h(q) = q$ requires $b(q) = 0$ in the interior. A stationary process with restoring drift generally fails this test. Equilibrium sampling and measurement absorption are different dynamical tasks.

Exercise 16. Complete record statistics For two equal-hazard channels with $q_1 + q_2 = 1$ and $\mathcal{A}(t) = t$, compute the waiting-time density, its mean, and the no-record probability at T. Repeat the survival calculation for $\mathcal{A}(t) = A_\infty(1 - e^{-t/t_0})$. Which case permits a permanent missed event?

18 An apparatus made from the fields

18.1 A pointer interaction already present in the response

In a homogeneous unit comoving volume, normalized response momenta are $P = a^3 \mathsf{G}_c \dot{y}$. The kinetic Hamiltonian is $H_{\rm kin} = P^T \mathsf{G}_c^{-1} P/(2a^3)$. Inverting Eq. (9.2) gives the cross term

$$H_{\rm cross} = \frac{\sinh(\phi/f)}{a^3\sqrt{c^2-1}\cosh^2(\phi/f)}\, P_U P_V. \tag{18.1}$$

Near the origin it is $\lambda_c \phi P_U P_V/a^3$. If P_U can be controlled over a finite experiment, and variations of a, pump depletion, and other interactions are bounded, this approximates

$$H_{\rm meas} = g(t)\phi P_V. \tag{18.2}$$

For an ideal short pulse with $\mathcal{G} = \int g(t)\mathrm{d}t$, Hamilton's equations give

$$\phi' = \phi, \qquad P_\phi' = P_\phi - \mathcal{G} P_V, \qquad V' = V + \mathcal{G}\phi, \qquad P_V' = P_V. \tag{18.3}$$

The pointer receives the signal and the system receives backreaction. The map is canonical. This is the same kinematic structure used in an ideal quantum measurement coupling, but it also exists classically.

If V is read with an independent initial width ΔV, the inferred signal error is approximately $\Delta V/|\mathcal{G}|$. The induced momentum disturbance has width $|\mathcal{G}|\Delta P_V$. Their product is $\Delta V \Delta P_V$. Classical preparation permits this product to be arbitrarily small. Quantum preparation supplies the bound $\Delta V \Delta P_V \geq \kappa/2$ only after the pointer's physical algebra and admissible states have been specified. Backreaction alone does not derive that bound.

18.2 From a displacement to a durable record

A pointer deflection is not yet a stable record. One possible apparatus module uses a coordinate Q in a double-well potential,

$$U(Q) = \frac{\lambda}{4}(Q^2 - v^2)^2 - j(t)Q. \tag{18.4}$$

The signal tilts the wells; relaxation can leave Q near one of the two stable values after the signal ends. The barrier is $\Delta U = \lambda v^4/4$ when $j = 0$. To calculate reliability, one must specify the bath, fluctuations, damping, preparation, and readout interval. Adding a friction term without an energy reservoir is not a closed Hamiltonian apparatus.

The research program's record models are valuable because they ask for these physical mechanisms rather than stopping at a normalized probability. They also expose where assumptions about thresholds, memory, equal gains, and environmental correlations enter the answer.

18.3 An internal clock must keep moving forward

A simple material clock is an oscillator with a nonzero phase-space radius. Suppose

$$\ddot{q} + 3H\dot{q} + \omega^2 q = 0 \tag{18.5}$$

with constant $\omega > 0$. Define an unwrapped phase by $\vartheta = \arg(\omega q - \mathrm{i}\dot{q})$, away from $q = \dot{q} = 0$. Differentiating the argument gives

$$\dot{\vartheta} = \omega \frac{\omega^2 q^2 + \dot{q}^2 + 3Hq\dot{q}}{\omega^2 q^2 + \dot{q}^2}. \tag{18.6}$$

Since $2\omega|q\dot{q}| \le \omega^2 q^2 + \dot{q}^2$,

$$\dot{\vartheta} \ge \omega - \frac{3}{2}|H|. \tag{18.7}$$

Thus $\omega > 3|H|/2$ throughout an interval is a sufficient condition for monotone phase on that interval. It is not a necessary condition, and a varying frequency or extra forces add terms.

Read another observable when this phase reaches a specified value, rather than at an externally stipulated coordinate time. This defines a relational reading. The clock's energy must still be included in the coupled dynamics; a physical clock does not become gravitationally free because we have called it a reference system.

18.4 Which operators does the apparatus actually measure?

A classical ensemble can be represented on the complex Hilbert space $L^2(\mathbb{R}^2, \mathrm{d}q\,\mathrm{d}p)$. The actual classical readings are multiplication operators $Q_{\mathrm{read}}\Psi = q\Psi$ and $P_{\mathrm{read}}\Psi = p\Psi$, which commute. The probability density $|\Psi|^2$ can be arbitrarily sharply concentrated in both variables.

On the same space, auxiliary differential operators

$$Q = \frac{q}{2} + \mathrm{i}\kappa\partial_p, \qquad P = \frac{p}{2} - \mathrm{i}\kappa\partial_q \tag{18.8}$$

obey $[Q, P] = \mathrm{i}\kappa I$ on a suitable common domain. That identity does not make them the measured variables. A physical coupling must identify which operator pair controls actual pointer statistics.

This is a precise version of the remaining quantum question. CR contains complex geometry, nontrivial symplectic reductions, interactions, and internal probes. To reconstruct quantum mechanics, one must show that preparable states and readable observables have the required noncommutative algebra and positivity, rather than merely displaying such an algebra in an auxiliary representation.

Exercise 17. The pointer map is symplectic Verify that Eq. (18.3) preserves $\mathrm{d}P_\phi \wedge \mathrm{d}\phi + \mathrm{d}P_V \wedge \mathrm{d}V$. Explain why this calculation is equally valid in classical mechanics.

Exercise 18. A clock bound Derive Eq. (18.7). For $H = 0$ and $q = A\cos(\omega\tau + \delta)$ with $A \neq 0$, show that the unwrapped phase advances exactly at rate ω. What fails at $A = 0$?

19 What would count as progress?

19.1 Four levels of a claim

It helps to classify a result before debating its significance.

Level	Example in these notes	What establishes it
Geometric theorem	The full real null cone forces $b = \chi g$.	An explicit proof from the stated premises.
Dynamical realization	The restricted curvature action equals an Einstein–scalar action.	A specified action, variables, boundaries, and a controlled variation.
Conditional prediction	A derivative-coupling ratio or a decay width.	The full model, state, normalization, kinematic regime, and corrections.
Physical identification	A response excitation is an observed particle or a quantum state.	A consistent observable interpretation and agreement with experimental tests.

The first three are already useful theoretical achievements. The fourth cannot be obtained by giving the first three a more ambitious title.

19.2 A short research program with concrete tests

For the foundations, determine which globally defined complex spacetimes and real slices support the local construction. For the critical sector, classify regular and singular branches and solve the reduced initial-value problem. For normal sectors, calculate scattering and loop counterterms to determine whether the tree-level relations survive renormalization or require additional symmetry.

For gauge structure, explain why the internal rank and splitting are selected, rather than merely realizing the group once they are chosen. For dark matter, diagonalize the full spectrum, include every allowed decay, and compute abundance and perturbations from specified initial conditions. For measurement, derive finite-resource preparation, pointer dynamics, record persistence, and sequential statistics from the same action.

The quantum task is particularly sharp: identify the physical observable algebra, establish positivity of all admissible states on that algebra, explain composition of systems, and determine the common action scale. Fitting a single qubit overlap is much less restrictive than reproducing entangled systems, incompatible measurements, and the statistics of actual apparatus.

19.3 The picture to keep

The framework begins with two foundations, complex spacetime and the real causal null cone. P1 relates them to clock-carrying observers and forces the induced metric into the form $\gamma = (1+\mathrm{i}\chi)g$. P2 connects natural complex momentum to the real mass shell and real canonical mechanics. A restricted curvature action turns the remaining geometric scalar into a canonical field. A chosen response action then produces positive, critical, and negative kinetic sectors, with curved normal target spaces and correlated interactions.

This is enough structure to calculate, and enough freedom to make model selection necessary. The most productive attitude is neither to dismiss every additional assumption nor to hide one. Write it down, derive what it buys, and find a test that could show it was the wrong choice. That is how a complex geometry becomes a physical research program.

A A compact mathematical toolkit

A.1 A metric is a bilinear rule, not just a matrix

A metric takes two tangent vectors and returns a scalar. Its matrix depends on a basis, but the number it assigns to the vectors does not. Under a real invertible change of basis $v = S\widetilde{v}$, a quadratic form transforms as $\widetilde{G} = S^T G S$. Positive definiteness and the numbers of positive, negative, and zero directions are invariant under such a transformation.

For a symmetric two-by-two matrix $G = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, positive definiteness is equivalent to $a > 0$ and $ad - b^2 > 0$. An equivalent criterion is positivity of both eigenvalues. For a positive matrix and a symmetric mass Hessian K, physical squared masses are the eigenvalues of $G^{-1/2} K G^{-1/2}$, or equivalently the roots of $\det(K - \mu G) = 0$. The latter form avoids choosing a particular square root in an algebraic calculation.

A.2 Derivatives and curvature

For a scalar, $\nabla_\mu \phi = \partial_\mu \phi$. For a vector,

$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu{}_{\mu\lambda} v^\lambda, \qquad \Gamma^\rho{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}).$$

Our Riemann convention is

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda} \Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda} \Gamma^\lambda{}_{\mu\sigma}, \qquad R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}.$$

These same formulas apply to a positive target metric when spacetime indices are replaced by field indices. The difference lies in the underlying space and signature, not in a new definition of curvature.

Normal coordinates set the connection to zero at one point. They cannot set a nonzero curvature tensor to zero. The familiar fact that a freely falling observer can remove local gravitational acceleration but not tidal effects has its field-space counterpart in Eq. (10.16): a local field redefinition can remove connection-dependent cubic terms while leaving invariant interactions.

A.3 Forms, canonical motion, and constraints

An exterior product is antisymmetric: $\mathrm{d}q \wedge \mathrm{d}p = -\mathrm{d}p \wedge \mathrm{d}q$. With $\Theta = p\,\mathrm{d}q$ and $\omega = \mathrm{d}\Theta = \mathrm{d}p \wedge \mathrm{d}q$,

$$\iota_{X_H} \omega = \dot{p}\,\mathrm{d}q - \dot{q}\,\mathrm{d}p = -\mathrm{d}H$$

gives $\dot{q} = \partial H/\partial p$ and $\dot{p} = -\partial H/\partial q$. For a constraint $C = 0$, a multiplier supplies its enforcement; requiring time preservation can generate additional constraints.

On a regular constraint surface, each independent first-class constraint, together with its gauge freedom, removes two phase-space dimensions. Each independent second-class constraint removes one. For N canonical pairs, F first-class constraints, and S second-class constraints, the configuration degree count is

$$N_{\text{physical}} = N - F - \frac{S}{2}.$$

This count assumes constant constraint rank and a properly completed constraint analysis. It must not be extrapolated through the singular surface $D = 0$ in Section 11.

A.4 Dimensions and the range of an effective theory

In four dimensions with $c_{\text{light}} = \hbar = 1$:

Quantity	Mass dimension
x^μ, proper time	-1
∂_μ, momentum, mass, f, $M_{\rm P}$	1
χ, F, ρ, c, $g_{\mu\nu}$, G_c	0
Canonical scalar ϕ, U, V	1
Lagrangian density and energy density	4
Derivative coefficient λ_c	-1
Spacetime curvature $R[g]$	2
Target scalar curvature in dimension-one field coordinates	-2

The operator $\phi\,\partial U\cdot\partial V$ has dimension five, so its coefficient has dimension minus one. The heavy-mediator operator in Eq. (9.8) has dimension eight and coefficient of dimension minus four. Such operators are organized in a low-energy expansion. Calling the full research program fundamental does not make every truncated action valid at arbitrarily high energy.

For fixed $c > 1$, amplitudes associated with target curvature grow schematically as E^2/f^2. Near $c = 1$, factors of $(c^2-1)^{-1}$ can reduce the useful perturbative range. A precise cutoff requires partial waves, the complete spectrum, and the relevant background. Dimensional analysis suggests where to look; it does not replace the unitarity calculation.

A.5 Why projective quantization gives $k+1$

The geometric-quantization statement in Section 16 has a simple algebraic part. Holomorphic sections of the degree-k line bundle on $\mathbb{CP}^1$ correspond to homogeneous polynomials $P(z_1, z_2)$ of total degree k. A basis is

$$z_1^k,\ z_1^{k-1}z_2,\ \ldots,\ z_2^k.$$

There are $k+1$ monomials. This explains the dimension once the quantization line bundle and its integer degree have been selected. It does not explain why physical dynamics should choose $k = 1$ or a particular value of the action quantum.

B Exercise solutions and checks

These answers give enough intermediate steps to audit the result. For a first reading, try each exercise before turning here; equations are much easier to recognize than to reproduce.

Exercise 1: cone versus scale

Since $\widetilde{g}(v,v) = 4g(v,v)$, the null sets agree exactly. For the same timelike displacement, $\mathrm{d}\widetilde{\tau} = \sqrt{4}\,\mathrm{d}\tau = 2\mathrm{d}\tau$. A positive conformal factor preserves the zero set but changes nonzero lengths. Light directions therefore determine a conformal class; clock calibration adds a scale.

Exercise 2: nine null tests

For $\boldsymbol{n} = \pm e_i$, subtraction gives $4b_{0i} = 0$ and addition gives $b_{ii} = -b_{00}$. For $\boldsymbol{n} = (e_i + e_j)/\sqrt{2}$, the equation reduces to $b_{ij} = 0$. Exactly one parameter, b_{00}, remains. The resulting matrix is $-b_{00}\eta$. An arbitrary function of direction can vanish at nine points and be nonzero elsewhere; the finite test is complete only because the model space is the finite-dimensional space of quadratic forms.

Exercise 3: the real constraint rank

Write $a = (1+\chi^2)^{-1}$. Then

$$\mathrm{d}\Re(F^{-1}C) = a\,\mathrm{d}C + C\,\mathrm{d}a, \qquad \mathrm{d}\Im(F^{-1}C) = -\chi a\,\mathrm{d}C - C\,\mathrm{d}(\chi a).$$

At $C = 0$ these become $a\,\mathrm{d}C$ and $-\chi a\,\mathrm{d}C$. They span a one-dimensional real subspace when $\mathrm{d}C \neq 0$. At a singular point with $\mathrm{d}C = 0$, this rank argument does not apply.

Exercise 4: a scalar-dependent mass

Variation of $-\int m\,\mathrm{d}\tau$ gives $u^\nu \nabla_\nu(mu_\mu) = -\partial_\mu m$. Expanding the left side and raising the index gives $m\,a^\mu = -\nabla^\mu m - u^\mu u^\nu \nabla_\nu m$. Divide by $m > 0$. Since $u_\mu u^\mu = -1$, contraction with u_μ vanishes. The force is the spatial projection of the mass gradient in the instantaneous rest frame. Constant mass removes it.

Exercise 5: rescaling a weighted field

The original equation is $\Box\varphi + 2\nabla s \cdot \nabla\varphi - m^2\varphi = 0$. Using $\varphi = e^{-s}\psi$,

$$\Box(e^{-s}\psi) = e^{-s}[\Box\psi - 2\nabla s \cdot \nabla\psi + ((\nabla s)^2 - \Box s)\psi].$$

The first-derivative terms cancel, leaving the equation stated in the exercise. The coefficient of $\Box\psi$ fixes the cone. The remaining background-dependent terms alter propagation at finite wavelength without changing that principal symbol.

Exercise 6: canonical normalization

Equate $\frac{1}{2}(\mathrm{d}\phi/\mathrm{d}\chi)^2 = 3M_{\mathrm{P}}^2/[4(1+\chi^2)]$. Choose a positive orientation and integrate with $\phi(0) = 0$: $\phi = \sqrt{3/2}M_{\mathrm{P}}\,\operatorname{arsinh}\chi$. The dimensions are $[\chi] = 0$, $[\phi] = [f] = [\partial_\mu] = 1$, and $[\mathcal{L}] = 4$. An analytic function such as sinh takes a dimensionless argument; ϕ/f supplies it.

Exercise 7: numerical eigenvalues

Here $\rho = 2$, so the ordered eigenvalues are $2(c+1), 2(c-1)$. For $c = 0, 1, 2, 3, 4$ they are $(2,-2), (4,0), (6,2), (8,4), (10,6)$, with determinants $-4, 0, 12, 32, 60$. For $\chi = \sqrt{3}$, a normalized positive eigenvector of K is $(\sqrt{3}/2, -1/2)^T$. Reversing χ reverses the sign of its second component. The eigenvalues and determinant are unchanged.

Exercise 8: an invariant amplitude

Insert the field redefinition into the three canonical kinetic terms and keep only first order in λ. The mixed terms from $-\frac{1}{2}(\partial\phi)^2$ cancel those proportional to $u\,\partial\varphi \cdot \partial v$ and $v\,\partial\varphi \cdot \partial u$ coming from the other two terms. The remaining contribution is $-\lambda\varphi\,\partial u \cdot \partial v$, which cancels the original interaction.

The mass terms change by

$$-\frac{1}{2}M^2\phi^2 - \frac{1}{2}m_U^2U^2 - \frac{1}{2}m_V^2V^2 \supset \frac{\lambda}{2}(M^2 - m_U^2 - m_V^2)\varphi uv.$$

This gives the same on-shell amplitude. A local invertible field redefinition changes the placement of the interaction among operators, not the tree-level physical decay amplitude when all induced terms are retained.

Exercise 9: shared-base curvature

Let $t_a = \operatorname{Tr} A_a = 2\tanh x/f$. The difference between the curvature of the full block metric and the sum of the individual curvatures is

$$-\frac{1}{4}\left[\left(\sum_a t_a\right)^2 - \sum_a t_a^2\right] = -\frac{k(k-1)}{f^2}\tanh^2 x.$$

For $c = 2, 3$, the coefficient of $\operatorname{sech}^2 x$ is $2 - \frac{1}{2}(4/3 + 9/8) = 37/48$ and the constant is -5. For $2, 3, 4$ at $x = 0$, $-13/6 - 33/16 - 61/30 = -501/80$. The extra shared-base term vanishes at the origin because $\tanh 0 = 0$, but generally not elsewhere.

Exercise 10: constraints and positivity

Differentiating Eq. (11.5) with respect to a_Y gives $2(N + a_Y D)$. Since $\partial a_Y/\partial Y = 1/(2f)$, preservation of $p_Y = 0$ gives $C_2 = 0$. The canonical bracket with p_Y differentiates with a minus sign, giving $-D/(2f)$. For $D > 0$, solve $a_Y = -N/D$ and substitute. The identity $|B - (A \cdot B)A/|A|^2|^2 = |B|^2 - (A \cdot B)^2/|A|^2$ proves positivity. Both solving the constraint and completing this particular square use $|A|^2 = D > 0$.

Exercise 11: exchange through the shared field

Differentiate E_y:

$$\dot{E}_y = \dot{y}^T G_c \ddot{y} + \frac{1}{2}\dot{\phi}\,\dot{y}^T G_c' \dot{y}.$$

The response equation gives $G_c\ddot{y} = -\dot{\phi}G_c'\dot{y}$, yielding the stated result. For homogeneous fields in signature $(-+++)$, $\Box\phi = -\ddot{\phi}$ and $\nabla y^i \cdot \nabla y^j = -\dot{y}^i\dot{y}^j$. Equation (12.2) therefore gives $\ddot{\phi} = \frac{1}{2}\dot{y}^T G_c' \dot{y}$ in this potential-free example. Thus $\frac{\mathrm{d}}{\mathrm{d}t}(\dot{\phi}^2/2) = -\dot{E}_y$.

Exercise 12: hypercharge sums

The exterior products have charges $-2/3$, $1/6$, and 1. For one left-handed generation the multiplicities and charges are $6(1/6)$, $3(-2/3)$, $3(1/3)$, $2(-1/2)$, and $1(1)$. Thus

$$\sum d_i Y_i = 1 - 2 + 1 - 1 + 1 = 0,$$

$$\sum d_i Y_i^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1 = 0.$$

These establish the mixed gravitational–hypercharge and cubic hypercharge cancellations for these assignments. The color and weak mixed anomaly coefficients also require their representation indices, while global questions depend on the group and spacetime structures.

Exercise 13: homogeneous free scalar

Let $a^3\dot{\phi} = C$. The Friedmann equation gives $H = |C|/(\sqrt{6}M_{\mathrm{P}}\,a^3)$ on the expanding branch. Hence $\mathrm{d}(a^3)/\mathrm{d}\tau = 3|C|/(\sqrt{6}M_{\mathrm{P}})$, so after choosing the time origin $a^3 \propto \tau$. It follows that $\dot{\phi} = \pm\sqrt{2/3}M_{\mathrm{P}}/\tau$, and integration gives the stated logarithm.

Multiplying only the scalar action by a constant preserves its source-free scalar equation but multiplies its stress tensor. Multiplying the entire coupled action by a common constant preserves all classical equations, while rescaling canonical normalization and the action relative to a fixed quantum $\hbar$. One must specify which operation is intended.

Exercise 14: the generalized eigenvalue

Let $r = \cosh x$. Since $\det G_4 = 15r^2$,

$$\det[\mathrm{diag}(5m_4^2, 0) - \mu G_4] = \mu[15\mu r^2 - 5m_4^2(4r - 1)].$$

The nonzero root is $m_4^2(4r - 1)/(3r^2)$. At $x = 0$ it equals m_4^2 and its first derivative vanishes, because r is even. The kinetic off-diagonal cubic interaction still exists; decay channels depend on the full spectrum and symmetry, not just this diagonal derivative.

Exercise 15: quantum positivity

The determinant is $e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta} - 2 = 2\cos\theta - 2 = -4\sin^2(\theta/2)$. The matrix must be positive for *every* finite test set. One special area with a vanishing determinant cannot repair the failure at other areas. The test concerns a state on the complete Weyl algebra, not only a selected commuting subset.

Exercise 16: waiting and missing

Let the common effective hazard be r_0, so $R = r_0$. For $\mathcal{A} = t$, survival is $e^{-r_0 t}$ and the waiting-time density is $r_0 e^{-r_0 t}$. Its mean is $1/r_0$, and the missed-event probability at T is $e^{-r_0 T}$. It vanishes as $T \to \infty$. For the saturating accumulator,

$$P_\varnothing(t) = \exp[-r_0 A_\infty(1 - e^{-t/t_0})], \qquad P_\varnothing(\infty) = e^{-r_0 A_\infty} > 0.$$

There is then an atom of probability at an infinite waiting time. The unconditional mean waiting time, counting permanent misses as infinite, is infinite.

Exercise 17: canonical backreaction

For a fixed pulse strength $\mathcal{G}$,

$$\begin{aligned} \mathrm{d}P'_\phi \wedge \mathrm{d}\phi' + \mathrm{d}P'_V \wedge \mathrm{d}V' &= (\mathrm{d}P_\phi - \mathcal{G}\mathrm{d}P_V) \wedge \mathrm{d}\phi \\ &\quad + \mathrm{d}P_V \wedge (\mathrm{d}V + \mathcal{G}\mathrm{d}\phi) \\ &= \mathrm{d}P_\phi \wedge \mathrm{d}\phi + \mathrm{d}P_V \wedge \mathrm{d}V. \end{aligned}$$

The cross terms cancel. Symplecticity is a property of Hamiltonian evolution; it does not constrain a classical ensemble to satisfy a nonzero minimum uncertainty product.

Exercise 18: monotone phase

Write $z = a + \mathrm{i}b$ with $a = \omega q$, $b = -\dot{q}$. Then $\dot{\vartheta} = (a\dot{b} - b\dot{a})/(a^2 + b^2)$. Using $\dot{b} = 3H\dot{q} + \omega^2 q$ gives the formula in the text. The inequality follows from $(\omega|q| - |\dot{q}|)^2 \geq 0$. For $H = 0$, $z = \omega A e^{\mathrm{i}(\omega\tau + \delta)}$ up to a constant phase when the amplitude sign changes, so the unwrapped phase rate is ω. At zero amplitude $z = 0$, its argument is undefined: no oscillating degree of freedom is available to serve as a phase clock.

C Notation, source map, and further reading

C.1 Keep the spaces separate

Symbol	Meaning
$M, \mathcal{Z}, \iota$	Real physical spacetime, complex ambient spacetime, and the embedding of the real slice.
g, G, γ	Real Lorentz metric, complex ambient metric, and induced complex metric $\gamma = \iota^* G$.
F, χ, ρ	Complex factor $1 + \mathrm{i}\chi$, real geometric coordinate, and modulus $\sqrt{1+\chi^2}$.
ϕ, f, x	Canonical geometric scalar, its normalization $f = \sqrt{3/2} M_{\mathrm{P}}$ in the chosen action, and $x = \phi/f$.
c, c_{light}	Dimensionless response parameter and physical light speed.
G_c, G_c	Response kinetic matrix and its reference-normalized form.
u, v, U, V	Original response coordinates and canonically normalized coordinates at the reference vacuum.
X, Y, w	Critical-sector coordinates and $w = \mathrm{d}X - Y\mathrm{d}\phi/(2f)$.
$h_{AB}, R_{\mathcal{T}}$	Metric and scalar curvature of the field-value target.
$p, P, C_{\mathrm{R}}, C_{\mathrm{A}}$	Real momentum, its natural complex extension, real shell, and compatible complex shell.
e	Real worldline multiplier imposing the mass shell.
$\mathcal{A}, \mathcal{F}$	Gauge connection and curvature when carrying spacetime/form indices; the explicitly time-dependent $\mathcal{A}(t)$ in Section 17 is accumulated work profile, an unrelated scalar function.
φ, ϱ	Reference matter scalar and quantum density operator.
κ	A proposed quantum action scale; set to physical $\hbar$ only after calibration.
H, τ, a	Hubble rate, cosmic proper time, and scale factor in cosmology; the matrix H_{ab} in curvature formulas is instead the response block.

C.2 Where the CR material comes from

The direct technical source for these notes is the revised synthesis listed as Ref. [1]. Its Sections 2–9 supply the geometric and particle foundations; Sections 10–15 supply free fields and the restricted gravitational action; Sections 16–18 and 27 supply response kinetics, target curvature, and interactions. Its Sections 20–32 contain the internal-bundle, gauge, dark-sector, and cosmological models. Sections 38–50 contain the projective, probabilistic, record, clock, and quantum-positivity investigations.

That synthesis collects the earlier geometry and c-sector manuscript, the integrated historical research reports, the P1/P2 v3.5 review, and the hundred-round comprehensive report. The present notes reorganize their results by prerequisite rather than research chronology. The examples, exercise sequence, intermediate explanations, and figures are pedagogical developments. They do not claim a new experimental confirmation or a new resolution of the open problems.

The source includes material beyond the scope of an introduction: microscopic composite spectra, detailed six-point amplitudes, higher-derivative and parent-action tests, four-form models, numerical apparatus studies, and the full historical index. Those calculations are better approached after the geometric and dynamical distinctions in this volume are secure.

C.3 Background reading

For a first encounter with manifolds, connections, and Einstein's equations, Carroll's introductory graduate notes provide a conventional reference.[2] Tong's general-relativity course covers the geometric machinery, curvature, and variational principles used here, with separate problem sheets.[3] For scalar field theory, Noether currents, quantization, and scattering, Tong's quantum-field theory notes provide the standard background against which the CR models should be compared.[4]

These resources support the established mathematical and physical tools. They are not independent sources for CR's particular postulates, response action, or proposed identification of sectors.

References

[1] CR research manuscripts, *CR Theory from P1 and P2: Complex Spacetime, the Common Null Cone, and Physical Dynamics*, revised synthesis, 2 November 2026. Source file: `CR_P1P2_Full_Theory_Reconstruction_English_2026-11-02.tex`. Author-provided research corpus and accompanying synthesis; the immediate source for the CR-specific results in these notes.

[2] S. M. Carroll, *Lecture Notes on General Relativity* (1997), arXiv:gr-qc/9712019.

[3] D. Tong, *General Relativity*, lecture notes and problem sheets, https://www.davidtong.org/teaching/general-relativity/.

[4] D. Tong, *Quantum Field Theory*, lecture notes, https://www.davidtong.org/teaching/quantum-field-theory/.